

HIGHER DIMENSIONAL MINIMAL MODEL PROGRAM FOR VARIETIES OF LOG GENERAL TYPE

CHRISTOPHER D. HACON

CONTENTS

1. Introduction	1
1.1. Notation	2
1.2. Preliminaries	4
1.3. Cone and Base Point Free Theorems	5
2. The main result	6
2.1. The minimal model program with scaling	9
2.2. Corollaries	10
3. Pl-flips	12
4. Multiplier ideal sheaves	13
4.1. Asymptotic multiplier ideal sheaves	16
4.2. Extending pluricanonical forms	18
5. Finite generation of the restricted algebra	22
5.1. Rationality of the restricted algebra	22
5.2. Proof of (3.9)	23
6. Log terminal models	24
6.1. Special termination	24
6.2. Existence of log terminal models	25
7. Non-vanishing	26
7.1. Nakayama-Zariski Decomposition	27
7.2. Non-vanishing	30
8. Finiteness of log terminal models	34
9. Solutions to some of the exercises	36
References	39

1. INTRODUCTION

Let X be a smooth complex projective variety. The minimal model program aims to show that if K_X is pseudo-effective (respectively if K_X is not pseudo-effective), then there exists a finite sequence

$$X = X_1 \dashrightarrow X_2 \dashrightarrow \dots \dashrightarrow X_n = \bar{X}$$

of well understood birational maps known as flips and divisorial contractions such that $\bar{X}$ is a minimal model i.e. $K_{\bar{X}}$ is nef (respectively $\bar{X}$ has the structure of a

Date: October 10, 2008.

The author was partially supported by NSF research grant no: 0456363 and an AMS Centennial fellowship.

Mori fiber space i.e. there is a morphism $f : \bar{X} \rightarrow Z$ such that $-K_{\bar{X}}$ is ample over Z).

The main features of the minimal model program had been understood in the 1980's by work of S. Mori, Y. Kawamata, J. Kollár, M. Reid, V. Shokurov and others. In order to produce the above sequence of birational maps $X_i \dashrightarrow X_{i+1}$, one proceeds as follows:

- (1) If K_{X_i} is nef, then X_i is a minimal model and there is nothing to do.
- (2) If K_{X_i} is not nef, then by the Cone Theorem, there exists a negative extremal ray and a corresponding morphism $f_i : X_i \rightarrow Z_i$ (of normal varieties, surjective with connected fibers and $\rho(X_i/Z_i) = 1$) such that $-K_{X_i}$ is ample over Z_i .
- (3) If $\dim Z_i < \dim X_i$, then $X_i \rightarrow Z_i$ is a Mori fiber space and we are done.
- (4) If $\dim Z_i = \dim X_i$ and $\dim \text{Ex}(f_i) = \dim X_i - 1$, then this is a divisorial contraction and we let $X_{i+1} = Z_i$.
- (5) If $\dim Z_i = \dim X_i$ and $\dim \text{Ex}(f_i) < \dim X_i - 1$, then this is a flipping contraction. In this case, Z_i has bad singularities and we may not let $X_{i+1} = Z_i$. Instead we let X_{i+1} be the flip of f_i cf. 3.1 (assuming that this exists).

In order to complete the minimal model program, it is therefore necessary to prove that flips exist and to show that there is no infinite sequence of flips. In dimension 3, the existence of flips was proved by S. Mori [Mori88], and the termination of flips is straightforward. In higher dimensions, the existence and termination of flips have proven to be difficult problems. Recently, V. Shokurov proved the existence of flips in dimension 4 (cf. [Shokurov03]) and then C. Hacon and J. McKernan (cf. [HM07]) showed that assuming the minimal model program in dimension $n - 1$ (for varieties of log-general type) then pl-flips exist in dimension n (cf. 3.3). Finally C. Birkar, P. Cascini, C. Hacon and J. McKernan (cf. [BCHM06]) showed that assuming that pl-flips exist in dimension n , then the minimal model program for varieties of log-general type holds in dimension n . In particular it follows that flips exist in all dimensions and that if X is a smooth complex projective variety, then its canonical ring $R(K_X) = \bigoplus_{m \geq 0} H^0(\mathcal{O}_X(mK_X))$ is finitely generated (this was also established for varieties of general type, using analytic methods, by Y.-T. Siu [Siu06]). We remark that in [BCHM06] it is shown that sequences of flips for the minimal model program with scaling terminate. This is sufficient to show that minimal models exist, however the problem of termination of an arbitrary sequence of flips remains open.

The main purpose of these lectures is to give a proof of the results of [HM07] and [BCHM06]. In particular we will show that flips exist (in all dimensions for any kawamata log terminal pair (X, Δ)) and that minimal models exist for any kawamata log terminal pair (X, Δ) such that $K_X + \Delta$ is big. In these lectures we will follow the approach of [HM08] of [BCHM06].

1.1. Notation. We work over the field of complex numbers $\mathbb{C}$.

If X is a normal variety, then $\text{WDiv}(X)$ will denote the group of Weil divisors on X . $\text{WDiv}_{\mathbb{R}}(X) = \text{WDiv}(X) \otimes_{\mathbb{Z}} \mathbb{R}$ is the group of $\mathbb{R}$ **divisors**. $D \in \text{WDiv}_{\mathbb{R}}(X)$ is $\mathbb{R}$ -**Cartier** if $D = \sum r_i D_i$ where $r_i \in \mathbb{R}$ and D_i are Cartier divisors. $D \in \text{WDiv}_{\mathbb{R}}(X)$ is **effective** if $D = \sum p_i P_i$ where P_i are prime divisors and $p_i \geq 0$. X is $\mathbb{Q}$ -**factorial**

if for any prime divisor $P \subset X$, there is a positive integer $m > 0$ such that mP is Cartier.

If $f : X \rightarrow U$ is a morphism of normal varieties and $D_i \in \text{WDiv}_{\mathbb{R}}(X)$, then $D_1 \sim_{\mathbb{R},U} D_2$ (i.e. D_1 and D_2 are **$\mathbb{R}$ -linearly equivalent over U**) if $D_1 - D_2 = \sum r_i(q_i) + f^*C$ where $r_i \in \mathbb{R}$, q_i are rational functions on X and C is an $\mathbb{R}$ -Cartier divisor on U . An $\mathbb{R}$ -Cartier divisor D is **nef over U** if $D \cdot C \geq 0$ for any curve $C \subset X$ contracted by f .

An $\mathbb{R}$ -Cartier divisor D is **ample over U** if $D \sim_{\mathbb{R},U} \sum r_i D_i$ where $r_i \in \mathbb{R}_{\geq 0}$ and D_i are ample Cartier divisors over U . An $\mathbb{R}$ -Cartier divisor D is **semiample over U** if there is a morphism $g : X \rightarrow Y$ over U such that $D \sim_{\mathbb{R},U} g^*D'$ where D' is ample over U . An $\mathbb{R}$ -Cartier divisor D is **big over U** if

$$\limsup \frac{\text{rk} f_* \mathcal{O}_X(mD)}{m^{\dim f}} > 0$$

where $\dim f$ denotes the dimension of a general fiber of f . Note that D is big over U if and only if $D \sim_{\mathbb{R},U} A + B$ where A is ample over U and $B \geq 0$. An $\mathbb{R}$ -Cartier divisor D is **pseudo-effective over U** if its numerical class belongs to the closure of the cone of big divisors over U .

The **real linear system** associated to an $\mathbb{R}$ -divisor D on X is

$$|D/U|_{\mathbb{R}} = \{C \geq 0 \mid C \sim_{\mathbb{R},U} D\}.$$

The **stable base locus** of D over U is the Zariski closed subset

$$\mathbf{B}(D/U) := \bigcap_{C \in |D/U|_{\mathbb{R}}} \text{Supp}(C).$$

When D is $\mathbb{Q}$ -Cartier then $\mathbf{B}(D/U)$ is the usual stable base locus. The **stable fixed divisor** of D over U is $\mathbf{Fix}(D/U)$ the divisorial component of $\mathbf{B}(D/U)$. The **augmented stable base locus** of D over U is given by $\mathbf{B}(D - \epsilon A/U)$ for any divisor A ample over U and any $0 < \epsilon \ll 1$. This set is denoted by $\mathbf{B}_+(D/U)$.

If D is a Weil divisor and $V \subset |D|$ is a linear series, then the **base locus of V** is

$$\text{Bs}(V) := \bigcap_{0 \leq C \in V} \text{Supp}(C)$$

and the **fixed divisor** $\text{Fix}(V)$ is the divisorial component of $\text{Bs}(V)$. We will abuse notation and we will denote $\text{Fix}(|D|)$ simply by $\text{Fix}(D)$. The divisor $\text{Mob}(D) = D - \text{Fix}(D)$ is the **mobile part of D** .

A **log pair** (X, Δ) is a normal quasi-projective variety X and an $\mathbb{R}$ -divisor $\Delta \geq 0$ such that $K_X + \Delta$ is $\mathbb{R}$ -Cartier.

A **log resolution** of a log pair (X, Δ) is a projective birational morphism $f : Y \rightarrow X$ from a smooth quasi-projective variety Y such that $f^{-1}(\Delta) \cup \text{Ex}(f)$ has simple normal crossings support. We will often write

$$(*) \quad K_Y + \Gamma = f^*(K_X + \Delta) + E$$

where K_Y is a canonical divisor such that $f_* K_Y = K_X$, Γ and E are effective divisors with no common components such that $f_* \Gamma = \Delta$ and $f_* E = 0$. The divisors Γ and E are uniquely determined.

We also write $E - \Gamma = \sum a_F(X, \Delta) F$ where F runs over all prime divisors in Y . For all but finitely many divisors $F \subset Y$, we have $a_F(X, \Delta) = 0$. The numbers $a_F(X, \Delta) = 0$ are the **discrepancies** of (X, Δ) along F . They are independent of the choice of f . We let $\text{disc.}(X, \Delta) = \min\{a_F(X, \Delta)\}$ for all divisors F over X .

If $a_F(X, \Delta) \geq -1$ (respectively $a_F(X, \Delta) > -1$) for all divisors F over X , then (X, Δ) is **log canonical** (respectively **kawamata log terminal**).

If $a_F(X, \Delta) \geq 0$ (respectively $a_F(X, \Delta) > 0$) for all exceptional divisors F over X , then (X, Δ) is **canonical** (respectively **terminal**).

If the coefficients of Δ are ≤ 1 and (X, Δ) admits a log resolution $f : Y \rightarrow X$ such that $a_F(X, \Delta) > -1$ for all exceptional divisors $F \subset Y$ over X , then (X, Δ) is **divisorially log terminal**. This is equivalent to requiring that there is a closed subset $Z \subset X$ such that $X - Z$ is smooth, $\Delta|_{X-Z}$ has simple normal crossings support and if F is an irreducible divisor with center contained in Z , then $a_F(X, \Delta) > -1$.

If $a_F(X, \Delta) > -1$ for all exceptional divisors F over X , then (X, Δ) is **purely log terminal**. If (X, Δ) is purely log terminal, then $\lfloor \Delta \rfloor = S$ is a disjoint union of normal prime divisors. If (X, Δ) is purely log terminal and $\lfloor \Delta \rfloor = S$ is irreducible, then the pair (S, Δ_S) defined by $K_S + \Delta_S = (K_X + \Delta)|_S$ is kawamata log terminal.

We will say that a $\mathbb{Q}$ -divisor A is a **general ample $\mathbb{Q}$ -divisor** if there is an integer $k > 0$ such that kA is a very ample Cartier divisor and $kA \in |kA|$ is general.

1.2. Preliminaries. We briefly recall several results that will be used in what follows.

Lemma 1.1 (Negativity Lemma). *Let $f : Y \rightarrow X$ be a proper birational morphism of normal quasi-projective varieties. If $-B$ is a f -nef $\mathbb{R}$ -Cartier divisor on Y , then*

- (1) $B \geq 0$ if and only if $h_* B \geq 0$.
- (2) If $B \geq 0$ and $x \in X$, then either $f^{-1}(y) \subset \text{Supp} B$ or $f^{-1}(y) \cap \text{Supp} B = \emptyset$.

Proof. See [KM98, 3.39]. □

Theorem 1.2 (Kawamata-Viehweg Vanishing). *If $f : Y \rightarrow X$ is a proper morphism of quasi-projective varieties with Y smooth, L is a line bundle on Y which is numerically equivalent to $M + \Delta$ where M is an f -nef and f -big $\mathbb{R}$ -Cartier divisor and (X, Δ) is kawamata log terminal. Then*

$$R^j f_* \mathcal{O}_Y(K_Y + L) = 0 \quad \text{for } j > 0.$$

Proof. See [KM98]. □

Corollary 1.3. *Let $f : Y \rightarrow X$ be a projective morphism of quasi-projective varieties with Y smooth, (Y, Γ) a divisorially log terminal pair and L a Cartier divisor such that $L - (K_Y + \Gamma)$ is f -ample, then*

$$R^j f_* \mathcal{O}_Y(L) = 0 \quad \text{for } j > 0.$$

Proof. Cf. (1.11). □

Theorem 1.4 (Kollár-Shokurov Connectedness). *Let $f : Y \rightarrow X$ be a proper birational morphism where Y is smooth and X is normal. If $D = \sum d_i D_i$ is an $\mathbb{R}$ -divisor with simple normal crossings support such that $f_* D \geq 0$ and $-(K_Y + D)$ is f -nef, then $\text{Supp}(\sum_{i: d_i \geq 1} D_i)$ is connected in a neighborhood of any fiber of f .*

Proof. See [KM98, 5.48]. □

Theorem 1.5 (Adjunction). *Let (X, Δ) be a log canonical pair and $S \subset \lfloor \Delta \rfloor$ a normal component of $\lfloor \Delta \rfloor$ of coefficient 1. Then there is an $\mathbb{R}$ -divisor Δ_S on S such that*

- (1) $(K_X + \Delta)|_S = K_S + \Delta_S$;
- (2) If (X, Δ) is divisorially log terminal then so is $K_S + \Delta_S$;
- (3) If (X, Δ) is purely log terminal, then $K_S + \Delta_S$ is kawamata log terminal;
- (4) If (X, S) is purely log terminal, then the coefficients of Δ_S at a codimension 1 point P have the form $(r-1)/r$ where r is the index of S at P (equivalently, r is the index of $K_X + S$ at P or r is the order of the cyclic group $\text{Weil}(\mathcal{O}_{X,P})$). It follows that if $\Delta = \sum \delta_i \Delta_i$, then $\text{mult}_P(\Delta|_S) = \sum \frac{m_i \delta_i}{r}$ where $m_i \in \mathbb{Z}_{\geq 0}$;
- (5) If (X, Δ) is purely log terminal, $f : Y \rightarrow X$ is a proper birational morphism of normal varieties $T = (f^{-1})_* S$ and $K_Y + \Gamma = f^*(K_X + \Delta)$, then $(f|_T)_*(E|_T) = 0$ and $(f|_T)_*(\Gamma_T) = \Delta_S$ where $(K_Y + \Gamma)|_T = K_T + \Gamma_T$.

1.3. Cone and Base Point Free Theorems. For the convenience of the reader, we recall the Cone and Base Point Free Theorems. We refer the reader to [KM98] for a particularly clear exposition.

Theorem 1.6 (Base Point Free Theorem). *Let $f : X \rightarrow Z$ be a proper morphism and (X, Δ) be a kawamata log terminal pair. If D is a f -nef Cartier divisor such that $aD - (K_X + \Delta)$ is f -nef and f -big for some $a > 0$, then $|bD|$ is f -free for all $b \gg 0$ (i.e. $f^*(f_* \mathcal{O}_X(D)) \rightarrow \mathcal{O}_X(D)$ is surjective).*

Theorem 1.7 (Cone Theorem). *Let $f : X \rightarrow Z$ be a projective morphism and (X, Δ) be a kawamata log terminal pair. Then*

- (1) *There exist countably many rational curves $C_j \subset X$ contracted by f such that $0 < -(K_X + \Delta) \cdot C_j \leq 2 \dim X$, and*

$$\overline{NE}(X/Z) = \overline{NE}(X/Z)_{(K_X + \Delta) \geq 0} + \sum \mathbb{R}_{\geq 0}[C_j].$$

- (2) *For any $\epsilon > 0$ and any f -ample divisor H ,*

$$\overline{NE}(X/Z) = \overline{NE}(X/Z)_{(K_X + \Delta + \epsilon H) \geq 0} + \sum_{\text{finite}} \mathbb{R}_{\geq 0}[C_j].$$

- (3) *If $F \subset \overline{NE}(X/Z)_{(K_X + \Delta) < 0}$ is a $(K_X + \Delta)$ negative extremal face, then there is a unique morphism over Z , $\text{cont}_F : X \rightarrow Z$ such that $(\text{cont}_F)_* \mathcal{O}_X = \mathcal{O}_Z$ and such that an irreducible curve $C \subset X$ is contracted by f if and only if $[C] \in F$.*
- (4) *If L is a line bundle on X such that $L \cdot C = 0$ for any curve C with $[C] \in F$, then $L = \text{cont}_F^* L_Z$ for some line bundle L_Z on Z .*

Recall that $N_1(X/Z) = \{\sum a_i C_i : a_i \in \mathbb{R}\} / \equiv_Z$ denotes the $\mathbb{R}$ vector space of 1-cycles modulo numerical equivalence over Z , that $NE(X/Z) = \{\sum a_i [C_i] : a_i \geq 0\} \subset N_1(X/Z)$, $\overline{NE}(X/Z)$ is the closure of $NE(X/Z)$ and that if D is an $\mathbb{R}$ -Cartier divisor, then $\overline{NE}(X/Z)_{D \geq 0} = \{x \in \overline{NE}(X/Z) : D \cdot x \geq 0\}$.

Often one says that $\overline{NE}(X/Z)$ is locally polyhedral even though this is not actually the case.

Exercise 1.8. Show that if $D \geq 0$ is a Cartier divisor, then there is an integer $m > 0$ such that $\mathbf{B}(D) = \mathbf{Bs}(mD)$.

Exercise 1.9. Show that if $f : Y \rightarrow X$ is a proper birational morphism of normal varieties and $D \geq 0$ is an $\mathbb{R}$ -divisor on X and G is an $\mathbb{R}$ -divisor on Y , then

- (1) If D is big (resp. pseudo-effective) then so is f^*D ;

- (2) If D is big (resp. pseudo-effective) and F is effective and its support contains all f -exceptional divisors, then $f_*^{-1}D + F$ is big (resp. pseudo-effective);
- (3) Show that if G is big (resp. pseudo-effective) then so is f_*G ;
- (4) Give an example where G is not big, but f_*G is big.

Exercise 1.10. Show that if (X, Δ) is a log pair and $a_F(X, \Delta) < -1$ for some divisor F over X , then $\min\{a_F(X, \Delta)\} = -\infty$.

Exercise 1.11. Show that (1.2) implies (1.3).

Exercise 1.12. Show that if (X, Δ) is a purely log terminal pair and $S = \lfloor D \rfloor$ is a prime divisor, then (S, Δ_S) is kawamata log terminal. Using the Connectedness Theorem cf. (1.4), show that if (S, Δ_S) is kawamata log terminal, then (X, Δ) is a purely log terminal pair on a neighborhood of S .

Exercise 1.13. Show that if (X, Δ) is a divisorially log terminal pair (resp. and $\mathbf{B}(\Delta)$ contains no logcanonical center of (X, Δ)), then there exists an $\mathbb{R}$ -divisor Δ_0 (resp. $\Delta_0 \sim_{\mathbb{R}} \Delta$) such that (X, Δ_0) is a kawamata log terminal pair. In particular, the log canonical centers of (X, Δ) are contained in $\text{Supp}(\Delta)$.

Exercise 1.14. Show that if (X, Δ) is a divisorially log terminal pair such that $K_X + \Delta$ is nef, $\Delta = A + B$ and $\mathbf{B}_+(A)$ contains no log canonical centers of (X, Δ) , then $K_X + \Delta$ is semiample. In particular if (X, Δ) is kawamata log terminal, $K_X + \Delta$ is nef and Δ is big, then $K_X + \Delta$ is semiample.

Exercise 1.15. Show that if (X, Δ) kawamata log terminal and $K_X + \Delta$ is big, then for any $0 < \epsilon \ll 1$, there is a kawamata log terminal pair (X, Δ') such that Δ' is big and $K_X + \Delta' \sim_{\mathbb{R}} (1 + \epsilon)(K_X + \Delta)$.

2. THE MAIN RESULT

Recall the following definitions.

Definition 2.1. Let $f : X \rightarrow Z$ be a proper morphism of normal quasi-projective varieties such that $f_*\mathcal{O}_X = \mathcal{O}_Z$ and $\rho(X/Z) = 1$. Let (X, Δ) be a log canonical pair such that $-(K_X + \Delta)$ is f -ample. If $\dim Z < \dim X$, then f is a **Mori fiber space**. If $\dim Z = \dim X$ and $\dim \text{Ex}(f) = \dim X - 1$, then f is a **divisorial contraction**. If $\dim Z = \dim X$ and $\dim \text{Ex}(f) < \dim X - 1$, then f is a **flipping contraction**.

Definition 2.2. Let $\pi : X \rightarrow U$ and $\pi' : Y \rightarrow U$ be projective morphisms of normal quasi-projective varieties. Let (X, Δ) be a divisorially log terminal pair and $\phi : X \dashrightarrow Y$ be a rational map over U such that

- (1) ϕ^{-1} contracts no divisors;
- (2) Y is $\mathbb{Q}$ -factorial;
- (3) $K_Y + \phi_*\Delta$ is nef over U ; and
- (4) $a(F, X, \Delta) < a(F, Y, \phi_*\Delta)$ for any ϕ -exceptional prime divisor $F \subset X$.

Then ϕ is a **log terminal model** for (X, Δ) .

Remark 2.3. Notice that by the Negativity Lemma, $a(F, X, \Delta) \leq a(F, Y, \phi_*\Delta)$ for any divisor F over Y . If $p : W \rightarrow X$ and $q : W \rightarrow Y$ are birational morphisms of normal varieties such that $\phi = q \circ p^{-1}$, then

$$p^*(K_X + \Delta) = q^*(K_Y + \phi_*\Delta) + E$$

where $E = \sum(a(F, Y, \phi_*\Delta) - a(F, X, \Delta)) \cdot F$ is effective and q -exceptional. If $k(K_X + \Delta)$ and $k(K_Y + \phi_*\Delta)$ are Cartier, then

$$H^0(\mathcal{O}_X(k(K_X + \Delta))) \cong H^0(\mathcal{O}_Y(k(K_Y + \phi_*\Delta))).$$

Notice that $\mathbf{Fix}(K_X + \Delta) = \text{Supp}(p_*E)$ is the set of all exceptional prime divisor $F \subset X$.

The main result of this paper is the following:

Theorem 2.4. *Let $\pi : X \rightarrow U$ be a projective morphism of normal quasi-projective varieties, (X, Δ) a $\mathbb{Q}$ -factorial kawamata log terminal pair such that Δ is big over U (or $K_X + \Delta$ is big over U or $K_X + \Delta$ is not pseudo-effective over U).*

Then there exists a finite sequence of flips and divisorial contractions over U

$$X = X_1 \dashrightarrow X_2 \dashrightarrow \dots \dashrightarrow X_n$$

such that either $K_{X_n} + \Delta_n$ is nef over U or $X_n \rightarrow Z$ is a Mori fiber space over U .

Proof. (2.14). □

Corollary 2.5. *Let (X, Δ) be a projective kawamata log terminal pair such that Δ (or $K_X + \Delta$) is big over U (resp. $K_X + \Delta$ is not pseudo-effective over U).*

Then (X, Δ) has a log terminal model over U (resp. there is a birational map over U $X \dashrightarrow \bar{X}$ such that $\bar{X} \rightarrow Z$ is a Mori fiber space over U).

Proof. Assume that Δ is big over U . Let $f : Y \rightarrow X$ be a log resolution and write

$$K_Y + \Gamma = f^*(K_X + \Delta) + E$$

as in (*) of §1.1. We may assume that there is an effective divisor F whose support contains the exceptional set. For $0 < \epsilon \ll 1$, $K_Y + \Gamma + \epsilon F$ is kawamata log terminal and $\Gamma + \epsilon F$ is big. By (2.4) there is a log terminal model $g : Y \dashrightarrow Y'$ for $K_Y + \Gamma + \epsilon F$ over U . It suffices to show that $\phi = g \circ f^{-1}$ is a log terminal model for $K_X + \Delta$ over U .

Let $p : W \rightarrow X$, $q : W \rightarrow Y$ and $r : W \rightarrow Y'$ be a common log resolution. Then

$$p^*(K_X + \Delta) + q^*(E + \epsilon F) = q^*(K_Y + \Gamma + \epsilon F) = r^*(K_{Y'} + g_*(\Gamma + \epsilon F)) + G$$

where G is effective and r -exceptional. Since $K_{Y'} + g_*(\Gamma + \epsilon F)$ is semiample over U cf. (1.14), it follows that $\mathbf{B}(r^*(K_{Y'} + g_*(\Gamma + \epsilon F)) + E/U) = \text{Supp}(G)$. Therefore, $X \dashrightarrow Y$ extracts no divisors and $g_*F = 0$. For any ϕ -exceptional divisor $P \subset X$, we have

$$a(P, X, \Delta) = a(P, Y, \Gamma + \epsilon F) < a(P, Y', \phi_*(\Gamma + \epsilon F)) = a(P, Y', \phi_*\Gamma).$$

Therefore $X \dashrightarrow Y$ is a log terminal model for (X, Δ) .

The case when $K_X + \Delta$ is big over U is similar. We may therefore assume that $K_X + \Delta$ is not pseudo-effective over U . Let A be an ample $\mathbb{Q}$ -divisor and $t > 0$ such that but $K_X + \Delta + tA$ is nef over U and kawamata log terminal. Let

$$\tau = \inf\{t > 0 \mid K_X + \Delta + tA \text{ is big}\}.$$

By what we have seen above, there exists a log terminal model $\phi : X' \rightarrow X$ for $(X, \Delta + \tau A)$ over X . In particular X' is $\mathbb{Q}$ -factorial and $X' \rightarrow X$ is a small birational morphism. If Δ' and A' denote the strict transforms of Δ and A , then $K_{X'} + \Delta' + A'$ is nef over U but not big over U . By (2.4) there is a finite sequence of flips and divisorial contractions over U $X' \dashrightarrow X'_n$ such that $X_n \rightarrow Z$ is a Mori fiber space over U . □

Corollary 2.6. *Let X be a smooth projective variety.*

Then $R(K_X)$ is finitely generated.

Proof. By [FM00], there exists a projective kawamata log terminal pair (Z, Δ_Z) such that $K_Z + \Delta_Z$ is big and $R(K_X)$ is finitely generated if and only if so is $R(K_Z + \Delta_Z)$. The Corollary now follows from (2.5), (1.15) and (1.14). $\square$

Theorem 2.7 (Theorem A: Existence of pl-flips). *Let (X, Δ) be a kawamata log terminal pair and $f : X \rightarrow Z$ be a pl-flipping contraction.*

Then the flip $f^+ : X^+ \rightarrow Z$ of f exists.

Theorem 2.8 (Theorem B: Existence of log terminal models). *Let $\pi : X \rightarrow U$ be a projective morphism of normal quasi-projective varieties. Let (X, Δ) be a kawamata log terminal pair and D be an effective $\mathbb{R}$ -divisor such that Δ is big over U and $K_X + \Delta \sim_{\mathbb{R}, U} D$.*

Then there exists a log terminal model for (X, Δ) over U .

Theorem 2.9 (Theorem C: Non-vanishing). *Let $\pi : X \rightarrow U$ be a projective morphism of normal quasi-projective varieties. Let (X, Δ) be a kawamata log terminal pair such that Δ is big over U .*

Then there exists an effective $\mathbb{R}$ -divisor D such that $K_X + \Delta \sim_{\mathbb{R}, U} D$.

Theorem 2.10 (Theorem D: Finiteness of models). *Let $\pi : X \rightarrow U$ be a projective morphism of normal quasi-projective varieties. Let $\mathcal{C} \subset \text{WDiv}_{\mathbb{R}}(X)$ be a rational polytope such that for any $K_X + \Delta \in \mathcal{C}$, Δ is big and (X, Δ) is kawamata log terminal.*

Then there exist finitely many $\phi_i : X \dashrightarrow Y_i$, $1 \leq i \leq k$, birational maps over U such that if $K_X + \Delta \in \mathcal{C}$ and $K_X + \Delta$ is pseudo-effective over U then

- (1) *There exists an index $1 \leq j \leq k$ such that ϕ_j is a log terminal model of (X, Δ) over U .*
- (2) *If $\phi : X \dashrightarrow Y$ is a log terminal model of (X, Δ) over U , then there exists an index $1 \leq j \leq k$ such that the rational map $\phi_j \circ \phi^{-1} : Y \dashrightarrow Y_j$ is an isomorphism.*

Theorem 2.11 (Theorem E: Zariski Decomposition). *Let $f : X \rightarrow Z$ be a projective morphism to a normal affine variety Z . Let (X, Δ) be a kawamata log terminal pair where $K_X + \Delta$ is pseudo-effective, $\Delta = A + B$ and $A \geq 0$ is an ample $\mathbb{Q}$ -divisor and $B \geq 0$. Then*

- (1) *(X, Δ) has a log terminal model $\mu : X \dashrightarrow Y$ over Z . In particular if $K_X + \Delta$ is $\mathbb{Q}$ -Cartier, then $R(K_X + \Delta)$ is finitely generated.*
- (2) *Let $V \subset \text{WDiv}_{\mathbb{R}}(X)$ be a finite dimensional affine subspace of $\text{WDiv}_{\mathbb{R}}(X)$ containing Δ which is defined over the rationals. There exists a constant $\delta > 0$ such that if P is a prime divisor contained in $\mathbf{B}(K_X + \Delta)$, then P is contained in $\mathbf{B}(K_X + \Delta')$, for any $\mathbb{R}$ -divisor $\Delta' \in V$ with $\|\Delta - \Delta'\| \leq \delta$.*
- (3) *Let $W \subset \text{WDiv}_{\mathbb{R}}(X)$ be the smallest affine subspace containing Δ which is defined over the rationals. Then there is a real number $\eta > 0$ and an integer $r > 0$ such that if $\Delta' \in W$, $\|\Delta - \Delta'\| \leq \eta$ and $k > 0$ is an integer such that $k(K_X + \Delta')/r$ is Cartier, then every component of $\text{Fix}(k(K_X + \Delta'))$ is a component of $\mathbf{B}(K_X + \Delta)$.*

Proof of Theorems A, B, C, D, E. The proof is by induction on the dimension $n = \dim X$. The case $n = 1$ is clear. We will show that Theorems A, B, C, D, E in

dimension $n - 1$ imply Theorem A in dimension n cf. (3.9), which implies Theorem B in dimension n cf. (6.1), which implies Theorem C in dimension n cf. (7.15), which implies Theorem D in dimension n cf. (8.1) and (8.2), which implies Theorem E in dimension n cf. (2.15). $\square$

2.1. The minimal model program with scaling. We start with a $\mathbb{Q}$ -factorial kawamata log terminal pair $(X, \Delta + H)$ and a projective morphism $\pi : X \rightarrow U$ such that $K_X + \Delta + H$ is nef over U and Δ is big over U . We let

$$\lambda = \inf\{t \geq 0 \mid K_X + \Delta + tH \text{ is nef over } U\}.$$

If $\lambda = 0$, then $K_X + \Delta$ is nef over U and we are done. If $\lambda > 0$, then there exists a $K_X + \Delta$ negative extremal ray R over U such that $(K_X + \Delta + \lambda H) \cdot R = 0$. We consider the corresponding contraction (over U) $\text{cont}_R : X \rightarrow Z$. If $\dim Z < \dim X$, then we have a $K_X + \Delta$ Mori fiber space and we are done. Otherwise (assuming the existence of the corresponding flips), we replace (X, Δ) by the corresponding flip or divisorial contraction $\phi : X \dashrightarrow X'$. Let $H' = \phi_* H$ and $\Delta' = \phi_* \Delta$. Since $K_X + \Delta + \lambda H$ is nef over U and $(K_X + \Delta + \lambda H) \cdot R = 0$, it follows that $K_{X'} + \Delta' + \lambda H'$ is nef over U and so we may repeat the process.

As in the usual minimal model program, this process terminates (yielding a log terminal model or a Mori fiber space) unless we get a sequence of flips $X_i \dashrightarrow X_{i+1}$. Let Δ_i and H_i denote the strict transforms of Δ and H on X_i . Then there is a sequence of real numbers $\lambda = \lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \dots > 0$ such that $K_{X_n} + \Delta_n + \lambda_n H_n$ is nef over U . In particular, $X \dashrightarrow X_n$ is a log terminal model for $(X, \Delta + \lambda_n H)$ over U . Notice that we have:

Remark 2.12. Throughout this paper, we will need the minimal model program with scaling in a slightly more context. We will start with $\pi : X \rightarrow U$ a projective morphism and $(X, \Delta + C)$ a divisorially log terminal $\mathbb{Q}$ -factorial pair such that $K_X + \Delta + C$ is nef over U , $\Delta = S + A + B$ where $\lfloor \Delta \rfloor = S$ and $\mathbf{B}_+(A/U)$ contains no log canonical centers of $(X, \Delta + C)$. Notice then that $K_X + \Delta \sim_{\mathbb{R}, U} K_X + \Delta'$ where (X, Δ') is kawamata log terminal and Δ' is big. Since any step of a $K_X + \Delta$ minimal model program over U is also a step of a $K_X + \Delta'$ minimal model program over U . Notice also that the condition that $\mathbf{B}_+(A/U)$ contains no log canonical centers of (X, Δ) is also preserved by any $K_X + \Delta$ minimal model program over U .

Lemma 2.13. $X_n \dashrightarrow X_m$ is not an isomorphism for any $m > n$.

Proof. Let E be a divisor over X whose center is contained in $\text{Ex}(\text{cont}_{R_n})$, where $\text{cont}_{R_n} : X_n \rightarrow Z_n$ is the corresponding extremal contraction over U . Then $a_E(X_n, \Delta_n) < a_E(X_{n+1}, \Delta_{n+1})$ and $a_E(X_{n+i}, \Delta_{n+i}) \leq a_E(X_{n+i+1}, \Delta_{n+i+1})$ for all $i > 0$ cf. (2.18) and (2.19). It follows that $a_E(X_n, \Delta_n) < a_E(X_m, \Delta_m)$ and so $X_n \dashrightarrow X_m$ is not an isomorphism.

Theorem 2.14. Let $(X, \Delta + H)$ be a $\mathbb{Q}$ -factorial kawamata log terminal pair and $\pi : X \rightarrow U$ a projective morphism such that $K_X + \Delta + H$ is nef over U and Δ is big over U .

Then any sequence of flips and divisorial contractions for the $K_X + \Delta$ minimal model program over U with scaling of H is finite.

Proof. By (2.10) and (2.13). $\square$

$\square$

Corollary 2.15. *Theorems 2.7 and 2.8 imply Theorem 2.11.*

Proof. (1) is immediate from (2.7) and (2.5). We now prove (2). Notice that if $K_X + \Delta'$ is not pseudo-effective, then $\mathbf{B}(K_X + \Delta') = X$. Therefore, we may assume that $K_X + \Delta'$ is pseudo-effective. Let $\phi : X \dashrightarrow Y$ be a log terminal model for $K_X + \Delta$ over Z . We may pick $0 < \delta$ such that if $\|\Delta - \Delta'\| \leq \delta$, then $K_Y + \phi_*\Delta'$ is kawamata log terminal cf. (8.4) and $a_F(X, \Delta') < a_F(Y, \phi_*\Delta')$ for any ϕ -exceptional divisor $F \subset X$. By (8.5), if $\psi : Y \rightarrow Z$ is a log terminal model for $K_Y + \phi_*\Delta'$ over Z , then $\psi \circ \phi$ is a log terminal model for $K_X + \Delta'$ over Z . By (2.21), all ϕ -exceptional divisors are contained in $\mathbf{B}(K_X + \Delta')$.

We now prove (3). Let $\phi : X \dashrightarrow Y$ be a log terminal model for $K_X + \Delta$ over Z and let $\mathcal{C}$ be the set of all $\Delta' \in W$ such that $\|\Delta - \Delta'\| \leq \eta$ and ϕ is a log terminal model for $K_X + \Delta'$ over Z . Since Δ is big, we may assume that there is $\mathbb{Q}$ -divisor A on Y ample over Z such that if $\|\Delta - \Delta'\| \leq \eta$, then $\phi_*\Delta' \geq A$. If η is sufficiently small, then $\phi_*(\Delta' - \Delta) + A/2$ is ample over Z for any $\Delta' \in W$ such that $\|\Delta - \Delta'\| \leq \eta$ and moreover

$$\mathcal{C} = \{\Delta' \in W \mid K_Y + \phi_*\Delta' \text{ is nef over } Z, \text{ and } \|\Delta - \Delta'\| \leq \eta\}.$$

By the Cone Theorem cf. (1.7), there are finitely many negative extremal rays $R_1, \dots, R_k$ for $K_Y + \phi_*\Delta - A/2$. Since

$$K_Y + \phi_*\Delta' = (K_Y + \phi_*\Delta - A/2) + \phi_*(\Delta' - \Delta) + A/2,$$

if $K_Y + \phi_*\Delta'$ is not nef over Z , then there is an integer $1 \leq i \leq k$ such that R_i is a negative extremal ray for $K_Y + \phi_*\Delta' - A/2$ over Z . Therefore $\mathcal{C}$ is cut out by a subset of the finite collection of extremal rays $R_i = \mathbb{R}^+[C_i]$ and hence $\mathcal{C}$ is a rational polytope. Therefore the affine subspace of $\text{WDiv}_{\mathbb{R}}(X)$ generated by $\mathcal{C}$ is W . We may then assume that $\mathcal{C} = \{\Delta' \in W : \|\Delta - \Delta'\| \leq \eta\}$.

Since Y is $\mathbb{Q}$ -factorial, there is an integer $l > 0$ such that if $G \in \text{WDiv}(Y)$, then lG is Cartier. By [Kollár93] cf. (1.6), there exists an integer $r > 0$ such that if $k(K_Y + \phi_*\Delta')/r$ is integral, then $k(K_Y + \phi_*\Delta')$ is generated by global sections. Let $p : W \rightarrow X$ and $q : W \rightarrow Y$ be a common resolution, then by definition of log terminal model, $p^*(K_X + \Delta') = q^*(K_Y + \phi_*\Delta') + E$ where $E \geq 0$ is q -exceptional and the support of $p_*(E)$ is the union of all ϕ -exceptional divisors on X . It follows that the support of $\text{Fix}(k(K_X + \Delta'))$ is given by the divisors contained in $\mathbf{B}(K_X + \Delta')$ i.e. by the divisors contained the support of $p_*(E)$. Since ϕ is a log terminal model for $K_X + \Delta$ over Z , these divisors are also contained in $\mathbf{B}(K_X + \Delta)$. $\square$

2.2. Corollaries.

Corollary 2.16. *Assume Theorems B, C, D in dimension n . Let $\pi : S \rightarrow U$ be a projective morphism of normal quasi-projective varieties, $(S, \Delta + C)$ be an n -dimensional kawamata log terminal pair and $S \dashrightarrow S_i$ be an infinite sequence of small birational maps of normal varieties over U . Let Δ_i and C_i be the strict transforms of Δ and C . If Δ is big over U and there are numbers $0 \leq \lambda_i \leq 1$ such that $K_{S_i} + \Delta_i + \lambda_i C_i$ is nef over U , then the set of rational maps $\phi_i : S \dashrightarrow S_i$ is finite modulo isomorphisms over U (i.e. there exist finitely many rational maps $\psi_j : S \dashrightarrow T_j$ over U such that for any i , there exists an integer j such that $\phi_i \circ \psi_j^{-1} : T_j \dashrightarrow S_i$ is an isomorphism)*

Proof. By (1.1), it follows that $(S_i, \Delta_i + \lambda_i C_i)$ is kawamata log terminal. Let $\nu_i : S'_i \rightarrow S_i$ be a log terminal model for $K_{S_i} + \Delta_i + \lambda_i C_i$ over S'_i . Then $\eta_i : S \dashrightarrow S'_i$ is a small birational morphism and so $K_{S'_i} + \Delta'_i + \lambda_i C'_i = \nu_i^*(K_{S_i} + \Delta_i + \lambda_i C_i)$ where Δ'_i and C'_i are the strict transforms of Δ_i and C_i . It follows that η_i is a log terminal model of $(S, \Delta + \lambda_i C)$ over U . By (2.10)_n, the set of rational maps $\eta_i : S \dashrightarrow S'_i$ is finite modulo isomorphisms over U . The map $\nu_i : S'_i \rightarrow S_i$ is given by the choice of a $K_{S'_i} + \Delta'_i + \lambda_i C'_i$ -trivial extremal face of $\overline{\text{NE}}(S'_i/U)$. Since Δ is big over U , $\Delta'_i + \lambda_i C'_i \sim_{\mathbb{R}, U} A + B$ where A is ample over U and $(S'_i, A + B)$ is kawamata log terminal. The above face is $K_{S'_i} + \frac{1}{2}A + B$ negative. By the Cone Theorem (1.7), there are finitely many $K_{S'_i} + \frac{1}{2}A + B$ negative extremal rays over U . Therefore, for any i , there are finitely many morphisms $S'_i \rightarrow S_i$. $\square$

Exercise 2.17. Show that if $f : X \rightarrow Z$ is a flipping contraction for a pair (X, Δ) , then $K_Z + f_*\Delta$ is not $\mathbb{Q}$ -Cartier.

Exercise 2.18. Let $f : X \rightarrow Z$ be a flipping contraction and $f^+ : X^+ \rightarrow Z$ be its flip. Show that X^+ is $\mathbb{Q}$ -factorial and $a_E(X, \Delta) \leq a_E(X^+, \Delta^+)$ for any divisor E over X and that strict inequality holds if the center of E is contained in the flipping or flipped locus. In particular, if (X, Δ) is log canonical, kawamata log terminal or divisorially log terminal, then so is (X^+, Δ^+) .

Exercise 2.19. Let $f : X \rightarrow Z$ be a divisorial contraction (i.e. f is a birational projective morphism of normal quasi-projective varieties such that X is $\mathbb{Q}$ -factorial, $\dim \text{Ex}(f) = \dim X - 1$, $\rho(X/Z) = 1$ and $-(K_X + \Delta)$ is f -ample). Show that Z is $\mathbb{Q}$ -factorial and $a_E(X, \Delta) \leq a_E(Z, f_*\Delta)$ for any divisor E over X and that strict inequality holds if the center of E is contained in $\text{Ex}(f)$. In particular, if (X, Δ) is log canonical, kawamata log terminal or divisorially log terminal, then so is $(Z, f_*\Delta)$.

Exercise 2.20. A $K_X + \Delta$ MMP is a sequence of flips and divisorial contractions $\phi_i : X_i \dashrightarrow X_{i+1}$ for (X_i, Δ_i) where $\Delta_i = (\phi_{i-1})_*\Delta_{i-1}$ and $(X_1, \Delta_1) = (X, \Delta)$. Use (2.18) and (2.19) to show that if Δ is a $\mathbb{Q}$ -divisor, then $R(K_X + \Delta) \cong R(K_{X_i} + \Delta_i)$.

Exercise 2.21. Let (X, Δ) be a kawamata log terminal pair and $\pi : X \rightarrow Z$ be a projective morphism such that Δ is big over Z . Show that the divisors contracted by a log terminal model of (X, Δ) over Z are the divisors contained in $\mathbf{B}(K_X + \Delta/Z)$.

Exercise 2.22. Let (X, Δ) be a kawamata log terminal pair and $X_i \dashrightarrow X_{i+1}$ a sequence of flips and divisorial contractions for the $K_X + \Delta$ minimal model program. If (X, Δ') is a kawamata log terminal pair such that $K_X + \Delta \equiv_{\mathbb{R}} K_X + \Delta'$, then show that $X_i \dashrightarrow X_{i+1}$ is a sequence of flips and divisorial contractions for the $K_X + \Delta'$ minimal model program.

Exercise 2.23. Let (X, Δ) be a kawamata log terminal pair. Let $\mathcal{E} = \{E_i\}$ be a finite collection of exceptional divisors over X such that $a(E, X, \Delta) < 0$. Show that there is a birational morphism $f : Y \rightarrow X$ from a $\mathbb{Q}$ -factorial variety such that $K_Y + \Gamma = f^*(K_X + \Delta)$ is kawamata log terminal, nef over X and the set of f -exceptional divisors $F \subset Y$ is $\mathcal{E}$. What can one say when (X, Δ) is not kawamata log terminal.

Exercise 2.24. Let (X, Δ) be a divisorially log terminal pair such that $\{\Delta\} = A + B$ where $\mathbf{B}_+(A)$ contains no log canonical center of (X, Δ) . Let $f : X \dashrightarrow X'$

be a flip or divisorial contraction and $\Delta' = f_*\Delta$, $A' = f_*A$. Show that $\mathbf{B}_+(A')$ contains no log canonical center of (X', Δ') .

3. PL-FLIPS

We begin by recalling the definition of a flip and of a pl-flip.

Definition 3.1. Let (X, Δ) be a log canonical pair. A **flipping contraction** is a small projective morphism $f : X \rightarrow Z$ of normal quasi projective varieties such that $f_*\mathcal{O}_X = \mathcal{O}_Z$, X is $\mathbb{Q}$ -factorial, $\rho(X/Z) = 1$ and $-(K_X + \Delta)$ is f -ample.

A small projective morphism $f^+ : X^+ \rightarrow Z$ of normal quasi-projective varieties is the **flip** of f if $f_*^+\mathcal{O}_{X^+} = \mathcal{O}_Z$, X^+ is $\mathbb{Q}$ -factorial, $\rho(X^+/Z) = 1$ and $K_{X^+} + \Delta^+$ is f -ample where Δ^+ denotes the strict transform of Δ .

Recall that a **small** morphism is a birational morphism such that $\dim \text{Ex}(f) < \dim X - 1$.

Remark 3.2. It is known that if the flip f^+ exists, then it is unique and given by

$$X^+ = \text{Proj}_Z \bigoplus_{m \geq 0} f_*\mathcal{O}_X(m(K_X + \Delta)).$$

To prove the existence of the flip f^+ , one may work locally over Z . If we assume that $Z = \text{Spec}(A)$ is affine and $\Delta \in \text{WDiv}_{\mathbb{Q}}(X)$, then it suffices to show that

$$R(K_X + \Delta) = \bigoplus_{m \geq 0} H^0(\mathcal{O}_X(m(K_X + \Delta)))$$

is a finitely generated A -algebra.

Definition 3.3. Let (X, Δ) be a purely log terminal pair with $S = \lfloor \Delta \rfloor$. A **pl-flipping contraction** is a flipping contraction $f : X \rightarrow Y$ such that $\Delta \in \text{WDiv}_{\mathbb{Q}}(X)$, S is irreducible and $-S$ is f -ample.

The flip of f is sometimes also called the **pl-flip** of f .

Remark 3.4. It is known that the existence of kawamata log terminal flips follows from the existence of pl-flips and special termination cf. [Fujino07]. Pl stands for pre-limiting.

Definition 3.5. Let (X, Δ) be a purely log terminal pair where $\lfloor \Delta \rfloor = S$ is irreducible. The **restricted algebra** $R_S(K_X + \Delta)$ is given by

$$\bigoplus_{m \geq 0} \text{Im} (H^0(\mathcal{O}_X(m(K_X + \Delta))) \rightarrow H^0(\mathcal{O}_S(m(K_X + \Delta)))) .$$

We have the following result due to V. Shokurov.

Theorem 3.6. *Let $f : X \rightarrow Z$ be a pl-flipping contraction for a purely log terminal pair (X, Δ) where $\lfloor \Delta \rfloor = S$. If $Z = \text{Spec}(A)$ is affine, then the flip of f exists if and only if the restricted algebra $R_S(K_X + \Delta)$ is finitely generated.*

Theorem (3.6) is useful because it allows us to proceed by induction on the dimension of X . Before proving (3.6), we will need some results about finite generation.

Lemma 3.7. *Let $R = \bigoplus_{m \geq 0} R_m$ be a graded ring and $d > 0$ be an integer. If R is an integral domain, then R is finitely generated if and only if so is $R^{(d)} = \bigoplus_{m \geq 0} R_{md}$.*

Proof. Suppose that R is finitely generated. Since $R^{(d)}$ is the ring of invariants of R by a $\mathbb{Z}/d\mathbb{Z}$ action, then $R^{(d)}$ is finitely generated by a Theorem of E. Noether cf. [Eisenbud95, 13.17].

Suppose that $R^{(d)}$ is finitely generated. Note that if $f \in R_m$, then f is integral over $R^{(d)}$ as it is a zero of the monic polynomial $x^d - f^d \in R^{(d)}[x]$. By E. Noether's Theorem on the finiteness of integral closures (cf. [Eisenbud95, 13.13]), it follows that R is finitely generated. $\square$

Proof of (3.6). If $R(K_X + \Delta) = \bigoplus_{m \geq 0} H^0(\mathcal{O}_X(m(K_X + \Delta)))$ is finitely generated, then clearly $R_S(K_X + \Delta)$ is finitely generated.

Assume now that $R_S(K_X + \Delta)$ is finitely generated. Let $S' \sim S$ be an effective divisor not containing S (cf. (3.11)) and let g be a rational function such that $S - S' = (g)$. There are positive integers a and b such that $a(K_X + \Delta) - bS' \equiv_Z 0$. By the Base Point Free Theorem cf. (1.6), $a(K_X + \Delta) - bS'$ is semiample over Z and so we may assume that $a(K_X + \Delta) \sim_Z bS'$. After replacing Z by an open subset, we may assume that $a(K_X + \Delta) \sim bS'$. By (3.7), $R(K_X + \Delta)$ is finitely generated if and only if so is $R(S')$. By (3.7), $R_S(S')$ is finitely generated and so it suffices to show that the kernel of $\phi : R(S') \rightarrow R_S(S')$ is finitely generated. Let g_m be a rational function corresponding to a non-zero element of $R(S')_m$ so that $(g_m) + mS' \geq 0$. If $\phi(g_m) = 0$, then $(g_m) + mS' = S + B$ for some $B \geq 0$. But then $(g_m/g) + (m-1)S' = B$ so that g_m/g is a rational function corresponding to a non-zero element of $R(S')_{m-1}$. In particular the kernel of ϕ is the principal ideal generated by g and the lemma follows. $\square$

In these lectures we will prove that a consequence of the existence of log terminal models for $(n-1)$ -dimensional log pairs of general type (cf. Theorem 2.11) implies the existence of pl-flips for log pairs of dimension n . Recall the following.

Definition 3.8. Let (X, Δ) be a kawamata log terminal pair and $f : X \rightarrow Z$ be a morphism to a normal variety. A **log terminal model** of (X, Δ) over Z is a birational morphism $\phi : X \dashrightarrow Y$ over Z such that ϕ^{-1} contracts no divisors, Y is normal and $\mathbb{Q}$ -factorial, $-(K_Y + \phi_*\Delta)$ is nef over Z and $a_E(X, \Delta) \leq a_E(Y, \phi_*\Delta)$ for all divisors E over X where the strict inequality holds if $E \subset X$ is a ϕ -exceptional divisor.

Theorem 3.9. Assume Theorem 2.11 in dimension $n-1$. Let $f : X \rightarrow Z$ be a pl-flipping contraction, then the flip $f^+ : X^+ \rightarrow Z$ exists.

Proof. See Lecture 5. $\square$

Exercise 3.10. Show that if $f : X \rightarrow Z$ is a flipping contraction for a divisorially log terminal pair (X, Δ) such that there is a component S of $\lfloor \Delta \rfloor$ where $-S$ is f -ample, then there is a pair (X, Δ') such that f is a pl-flipping contraction for (X, Δ') .

Exercise 3.11. Let $f : X \rightarrow Z$ be a flipping contraction where Z is affine. Show that for any irreducible divisor $S \subset X$ we have $S \sim S'$ where S' is an effective divisor whose support does not contain S .

4. MULTIPLIER IDEAL SHEAVES

In this section we will recall the definition and the main properties of multiplier ideal sheaves. The standard reference for multiplier ideal sheaves is [Lazarsfeld04].

In what follows we will focus on a generalization of this notion known as adjoint ideals.

Definition 4.1. Let X be a smooth quasi-projective variety, Δ be a reduced divisor with simple normal crossings support and V a linear series such that $\text{Bs}(V)$ contains no log canonical center of (X, Δ) (i.e. $\text{Bs}(V)$ contains no strata of Δ). Let $f: Y \rightarrow X$ be a log resolution of V and of (X, Δ) . We let

$$K_Y + \Gamma = f^*(K_X + \Delta) + E$$

as in (*) of §1.1.

For any real number $c > 0$, we define the **multiplier ideal sheaf**

$$\mathcal{J}_{\Delta, c, V} := f_* \mathcal{O}_Y(E - \lfloor cF \rfloor)$$

where $F = \text{Fix}(f^*V)$.

If $D \geq 0$ is a $\mathbb{Q}$ -divisor and $m > 0$ is an integer such that mD is Cartier, then we let $\mathcal{J}_{\Delta, c, D} = \mathcal{J}_{\Delta, \frac{c}{m}, V}$ where $V = \{mD\}$ is the linear series consisting of the unique divisor mD .

Remark 4.2. If $\Delta = 0$, then $\mathcal{J}_{\Delta, c, V} = \mathcal{J}_{c, V}$ is the usual multiplier ideal sheaf.

Lemma 4.3. *The definition given in (4.1) is independent of the log resolution.*

Proof. Let $f: Y \rightarrow X$ and $f': Y' \rightarrow X'$ be two log resolution. Since any two log resolutions can be dominated by a third log resolution, we may assume that there is a morphism $\mu: Y' \rightarrow Y$ such that $f' = f \circ \mu$. We let

$$K_Y + \Gamma = f^*(K_X + \Delta) + E \quad \text{and} \quad K_{Y'} + \Gamma' = (f')^*(K_X + \Delta) + E'$$

be as in (*) of §1.1. We have

$$K_{Y'} + \Gamma' = \mu^*(K_Y + \Gamma - E) + E'.$$

Let $F = \text{Fix}(f^*V)$ and $F' = \text{Fix}((f')^*V)$. Since $f^*V - F$ is base point free, it follows that $F' = \mu^*F$. We claim that

$$\mu_* \mathcal{O}_{Y'}(E' - \mu^*E - \lfloor \mu^*\{cF\} \rfloor) = \mathcal{O}_Y.$$

Grant this for the time being, then

$$\begin{aligned} (f')_* \mathcal{O}_{Y'}(E' - \lfloor cF' \rfloor) &= (f')_* \mathcal{O}_{Y'}(E' - \mu^*E - \lfloor \mu^*\{cF\} \rfloor + \mu^*(E - \lfloor cF \rfloor)) = \\ &= f_* (\mu_* \mathcal{O}_{Y'}(E' - \mu^*E - \lfloor \mu^*\{cF\} \rfloor) \otimes \mathcal{O}_Y(E - \lfloor cF \rfloor)) = f_* \mathcal{O}_Y(E - \lfloor cF \rfloor). \end{aligned}$$

The Lemma now follows immediately. To see the claim, notice that $K_Y + \Gamma + \{cF\}$ is divisorially log terminal and its log canonical places coincide with those of $K_Y + \Gamma - E$. From the equation

$$K_{Y'} + \Gamma' + \mu^*(E + \{cF\}) - E' = \mu^*(K_Y + \Gamma + \{cF\})$$

it follows that $\lfloor \mu^*(E + \{cF\}) - E' \rfloor \leq 0$ so that $E' - \mu^*E - \lfloor \mu^*\{cF\} \rfloor$ is effective and μ -exceptional. The claim is now clear. $\square$

Lemma 4.4. *Let X be a smooth quasi-projective variety, Δ be a reduced divisor with simple normal crossings support and $V \subset V'$ be linear series such that $\text{Bs}(V)$ contains no log canonical center of (X, Δ) . Let $c \geq c'$ be positive real numbers, $\Delta' \leq \Delta$ be reduced divisors and D and D' be effective $\mathbb{Q}$ -Cartier divisors whose supports contain no log canonical centers of (X, Δ) .*

Then

- (1) $\mathcal{J}_{\Delta, c, V} \subset \mathcal{J}_{\Delta', c', V'}$, and in particular $\mathcal{J}_{\Delta, c, V} \subset \mathcal{J}_{c, V} \subset \mathcal{O}_X$.

- (2) If Σ is an effective Cartier divisor such that $D \leq \Sigma + D'$ and $\mathcal{J}_{\Delta, D'} = \mathcal{O}_X$, then $\mathcal{I}_\Sigma \subset \mathcal{J}_{\Delta, D}$.

Proof. Let $f: Y \rightarrow X$ be a common log resolution of V , V' , (X, Δ) and (X', Δ') . We let

$$K_Y + \Gamma = f^*(K_X + \Delta) + E \quad \text{and} \quad K_Y + \Gamma' = f^*(K_X + \Delta') + E'$$

be as in (*) of §1.1. We have that $\Gamma' \leq \Gamma$ and $E' \geq E$. If we let $F = \text{Fix}(f^*V)$ and $F' = \text{Fix}(f^*V')$, then we also have $F \geq F'$. It follows that

$$\mathcal{J}_{\Delta, c \cdot V} = f_* \mathcal{O}_Y(E - \lfloor cF \rfloor) \subset f_* \mathcal{O}_Y(E' - \lfloor c'F' \rfloor) = \mathcal{J}_{\Delta', c' \cdot V'}.$$

Let $f: Y \rightarrow X$ be a log resolution of $(X, \Delta + \Sigma + D')$. We let

$$K_Y + \Gamma = f^*(K_X + \Delta) + E$$

be as in (*) of §1.1. Since $\mathcal{J}_{\Delta, D'} = \mathcal{O}_X$, we have $E - \lfloor f^*D' \rfloor \geq 0$. Since Σ is Cartier, we have $\lfloor f^*D \rfloor \leq f^*\Sigma + \lfloor f^*D' \rfloor$. Therefore

$$E - \lfloor f^*D \rfloor \geq E - \lfloor f^*D' \rfloor - f^*\Sigma \geq -f^*\Sigma$$

and hence

$$\mathcal{I}_\Sigma = f_* \mathcal{O}_Y(-f^*\Sigma) \subset f_* \mathcal{O}_Y(E - \lfloor f^*D \rfloor) = \mathcal{J}_{\Delta, D}.$$

□

The next lemma shows that multiplier ideals of the form $\mathcal{J}_{\Delta, D}$ may be viewed as being obtained by successive extensions of the usual multiplier ideal sheaves $\mathcal{J}_D$.

Lemma 4.5. *Let X be a smooth quasi-projective variety, Δ be a reduced divisor with simple normal crossings support and $D \geq 0$ be a $\mathbb{Q}$ -Cartier divisor whose supports contains no log canonical center of (X, Δ) . Let S be a component of Δ , then there is a short exact sequence*

$$0 \rightarrow \mathcal{J}_{\Delta-S, D+S} \rightarrow \mathcal{J}_{\Delta, D} \rightarrow \mathcal{J}_{(\Delta-S)|_S, D|_S} \rightarrow 0.$$

Proof. Let $f: Y \rightarrow X$ be a log resolution of $(X, \Delta + D)$ which is an isomorphism at the generic point of each log canonical center of (X, Δ) (cf. [Szab694]). We write

$$K_Y + \Gamma = f^*(K_X + \Delta) + E$$

as in (*) of §1.1. Let $T = (f^{-1})_* S$ and consider the short exact sequence

$$0 \rightarrow \mathcal{O}_Y(E - \lfloor f^*D \rfloor - T) \rightarrow \mathcal{O}_Y(E - \lfloor f^*D \rfloor) \rightarrow \mathcal{O}_T(E - \lfloor f^*D \rfloor) \rightarrow 0.$$

By definition, we have $f_* \mathcal{O}_Y(E - \lfloor f^*D \rfloor) = \mathcal{J}_{\Delta, D}$. Since

$$K_Y + \Gamma - T = f^*(K_X + \Delta - S) + E + f^*S - T$$

where $\Gamma - T$ and $E + f^*S - T$ are effective with no common components, it follows that

$$\mathcal{J}_{\Delta-S, D+S} = f_* \mathcal{O}_Y(E + f^*S - T - \lfloor f^*(D + S) \rfloor) = f_* \mathcal{O}_Y(E - \lfloor f^*D \rfloor - T).$$

Since

$$K_T + (\Gamma - T)|_T = (f|_T)^*(K_S + (\Delta - S)|_S) + E|_T,$$

it follows that

$$\mathcal{J}_{(\Delta-S)|_S, D|_S} = (f|_T)_* \mathcal{O}_T(E - \lfloor f^*D \rfloor).$$

Finally, pick an effective and f -exceptional $\mathbb{Q}$ -divisor F such that $-F$ is f -ample and $(Y, \Gamma - T + \{f^*D\} + F)$ is divisorially log terminal cf. (4.20). Since

$$E - \lfloor f^*D \rfloor - T = K_Y + \Gamma - T + \{f^*D\} + F - F - f^*(K_X + \Delta + D),$$

by Kawamata-Viehweg vanishing (cf. (1.2)), we have that $R^1 f_* \mathcal{O}_Y(E - \lfloor f^* D \rfloor - T) = 0$. $\square$

Lemma 4.6. *Let X be a smooth quasi-projective variety, Δ be a reduced divisor with simple normal crossings support and $D \geq 0$ be a $\mathbb{Q}$ -Cartier divisor whose support contains no log canonical center of (X, Δ) . Let $\pi: X \rightarrow Z$ be a projective morphism to a normal affine variety Z and N be a Cartier divisor on X such that $N - D$ is ample. Then*

$$H^i(X, \mathcal{J}_{\Delta, D}(K_X + \Delta + N)) = 0 \quad \text{for } i > 0,$$

and if S is a component of Δ , then the homomorphism

$$H^0(X, \mathcal{J}_{\Delta, D}(K_X + \Delta + N)) \rightarrow H^0(S, \mathcal{J}_{(\Delta - S)|_S, D|_S}(K_X + \Delta + N))$$

is surjective.

Proof. Let $f: Y \rightarrow X$ be a log resolution of $(X, \Delta + D)$ which is an isomorphism at the generic point of each log canonical center of (X, Δ) (cf. [Szabó94]). We write

$$K_Y + \Gamma = f^*(K_X + \Delta) + E$$

as in (*) of §1.1. Pick an effective and f -exceptional $\mathbb{Q}$ -divisor F such that $f^*(N - D) - F$ is ample and $(Y, \Gamma + \{f^* D\} + F)$ is divisorially log terminal. Since

$$E - \lfloor f^* D \rfloor + f^*(K_X + \Delta + N) = K_Y + \Gamma + \{f^* D\} + F + f^*(N - D) - F,$$

by Kawamata-Viehweg vanishing (cf. (1.2)), we have that $R^j f_* \mathcal{O}_Y(E - \lfloor f^* D \rfloor) = 0$ for all $j > 0$. Since $f_* \mathcal{O}_Y(E - \lfloor f^* D \rfloor) = \mathcal{J}_{\Delta, D}$, by Kawamata-Viehweg vanishing it follows then that

$$H^i(X, \mathcal{J}_{\Delta, D}(K_X + \Delta + N)) = H^i(Y, \mathcal{O}_Y(E - \lfloor f^* D \rfloor + f^*(K_X + \Delta + N))) = 0$$

for all $i > 0$ cf. (4.21).

The final claim follows immediately from (4.5). $\square$

4.1. Asymptotic multiplier ideal sheaves.

Definition 4.7. Let D be a divisor on a normal variety X and $V_i \subset |iD|$ be linear series, then $V_\bullet$ is an **additive sequence of linear series** if

$$V_i + V_j \subset V_{i+j}$$

for all $i, j \geq 0$.

Lemma 4.8. *Let X be a smooth quasi-projective variety and Δ be a reduced divisor with simple normal crossings support. If $V_\bullet$ is an additive sequence of linear series and c is a positive real number, k a positive integer such that $\text{Bs}(V_k)$ contains no log canonical centers of (X, Δ) , then*

$$\mathcal{J}_{\Delta, \frac{c}{p} \cdot V_p} \subset \mathcal{J}_{\Delta, \frac{c}{q} \cdot V_q}$$

for any integers p and q such that k divides p and p divides q .

In particular, since X is Noetherian, the set $\{\mathcal{J}_{\Delta, \frac{c}{p} \cdot V_p}\}$ has a unique maximal element $\mathcal{J}_{\Delta, \|c \cdot V\|}$. Note that $\mathcal{J}_{\Delta, \|c \cdot V\|} = \mathcal{J}_{\Delta, \frac{c}{p} \cdot V_p}$ for any $p > 0$ sufficiently divisible.

Proof. This follows from (1) of (4.4), since $(q/p)V_p \subset V_q$. $\square$

Definition 4.9. Let X be a smooth quasi-projective variety, Δ be a reduced divisor with simple normal crossings support and S an irreducible component of Δ . If c is a positive real number, D is a $\mathbb{Q}$ -divisor on X such that $\mathbf{B}(D)$ contains no log canonical centers of (X, Δ) then we define the **asymptotic multiplier ideal sheaves**

$$\mathcal{J}_{\Delta, c \cdot \|D\|} = \mathcal{J}_{\Delta, c \cdot V_\bullet} \quad \text{and} \quad \mathcal{J}_{(\Delta-S)|_S, c \cdot \|D\|_S} = \mathcal{J}_{(\Delta-S)|_S, c \cdot W_\bullet}$$

where $V_m = |mD|$ and $W_m = |mD|_S$ is the restriction (or trace) of the linear series $|mD|$ to S .

Lemma 4.10. *Let X be a smooth quasi-projective variety, Δ be a reduced divisor with simple normal crossings support, S an irreducible component of Δ and D a $\mathbb{Q}$ -divisor on X such that $\mathbf{B}(D)$ contains no log canonical centers of (X, Δ) . If $\pi: X \rightarrow Z$ is a projective morphism to a normal affine variety. Then*

- (1) *For any real numbers $0 < c' \leq c$ and any reduced divisor $0 \leq \Delta' \leq \Delta$, we have*

$$\mathcal{J}_{\Delta, \|c \cdot D\|} \subset \mathcal{J}_{\Delta', \|c' \cdot D\|}.$$

- (2) *If D is Cartier, then*

$$\text{Im}(\pi_* \mathcal{O}_X(D) \rightarrow \pi_* \mathcal{O}_S(D)) \subset \pi_* \mathcal{J}_{(\Delta-S)|_S, \|D\|_S}(D).$$

- (3) *If D is Cartier, A is an ample Cartier divisor and H is a very ample Cartier divisor, then*

$$\mathcal{J}_{\|D\|}(K_X + D + A + nH)$$

is globally generated where $n = \dim X$.

- (4) *If D is Cartier and $\mathbf{B}_+(D)$ contains no log canonical centers of (X, Δ) , then the image of the homomorphism*

$$\pi_* \mathcal{O}_X(K_X + \Delta + D) \rightarrow \pi_* \mathcal{O}_S(K_X + \Delta + D)$$

contains

$$\pi_* \mathcal{J}_{(\Delta-S)|_S, \|D\|_S}(K_X + \Delta + D).$$

Proof. (1) is immediate from (1) of (4.4).

To see (2), let $p > 0$ be an integer such that $\mathcal{J}_{(\Delta-S)|_S, \|D\|_S} = \mathcal{J}_{(\Delta-S)|_S, \frac{1}{p} \cdot |pD|_S}$ and consider a log resolution $f: Y \rightarrow X$ of (X, Δ) and $|pD|$. We let

$$K_Y + \Gamma = f^*(K_X + \Delta) + E$$

be as in (*) of §1.1. Let $F_p = \text{Fix}(f^*|pD|)$, then $pF_1 \geq F_p$ so that

$$\pi_* \mathcal{O}_X(D) = (\pi \circ f)_* \mathcal{O}_Y(f^*D - F_1) \subset (\pi \circ f)_* \mathcal{O}_Y(f^*D + E - \lfloor \frac{1}{p} F_p \rfloor)$$

$$\subset (\pi \circ f)_* \mathcal{O}_Y(f^*D + E) = \pi_* \mathcal{O}_X(D).$$

The assertion now follows as

$$\pi_* \mathcal{J}_{(\Delta-S)|_S, \|D\|_S}(D) = (\pi \circ f)_* \mathcal{O}_T(f^*D + E - \lfloor \frac{1}{p} F_p \rfloor)$$

where $T = (f^{-1})_* S$.

- (3) is immediate from (4.6) and (4.22).

To see (4), let $p > 0$ be an integer such that $pD \sim A + B$ where A is a general very ample divisor and B is an effective divisor containing no log canonical centers of (X, Δ) . Replacing p by a multiple, we may assume that $\mathcal{J}_{\Delta, \|D\|_S} = \mathcal{J}_{\Delta, \frac{1}{p} \cdot |pD|_S}$.

Let $f: Y \rightarrow X$ be a log resolution of (X, Δ) and $|pD|$ which is an isomorphism at the generic point of each log canonical center of (X, Δ) . We let

$$K_Y + \Gamma = f^*(K_X + \Delta) + E$$

be as in (*) of §1.1. Let $F_p = \text{Fix}(f^*|pD|)$, then $M_p := pf^*D - F_p$ is base point free and $f^*B \geq F_p$. Let $T = (f^{-1})_*S$ and F be an effective and exceptional divisor such that $f^*A - F$ is ample and $\delta > 0$ be a rational number such that $K_Y + \Gamma - T + \{\frac{1}{p}F_p\} + \delta(f^*B - F_p + F)$ is divisorially log terminal. We have

$$E - \lrcorner \frac{1}{p}F_p \lrcorner - T + f^*(K_X + \Delta + D) = K_Y + \Gamma - T + f^*D - \lrcorner \frac{1}{p}F_p \lrcorner \sim_{\mathbb{Q}}$$

$$K_Y + \Gamma - T + \{\frac{1}{p}F_p\} + \delta(f^*B - F_p + F) + (\frac{1}{p} - \delta)M_p + \delta(f^*A - F).$$

By Kawamata-Viehweg vanishing,

$$R^1(\pi \circ f)_* \mathcal{O}_Y(E - \lrcorner \frac{1}{p}F_p \lrcorner - T + f^*(K_X + \Delta + D)) = 0.$$

Therefore, pushing forward the short exact sequence

$$\begin{aligned} 0 \rightarrow \mathcal{O}_Y(E - \lrcorner \frac{1}{p}F_p \lrcorner - T + f^*(K_X + \Delta + D)) &\rightarrow \mathcal{O}_Y(E - \lrcorner \frac{1}{p}F_p \lrcorner + f^*(K_X + \Delta + D)) \\ &\rightarrow \mathcal{O}_T(E - \lrcorner \frac{1}{p}F_p \lrcorner + f^*(K_X + \Delta + D)) \rightarrow 0, \end{aligned}$$

one sees that the homomorphism

$$(\pi \circ f)_* \mathcal{O}_Y(E - \lrcorner \frac{1}{p}F_p \lrcorner + f^*(K_X + \Delta + D)) \rightarrow (\pi \circ f)_* \mathcal{O}_T(E - \lrcorner \frac{1}{p}F_p \lrcorner + f^*(K_X + \Delta + D))$$

is surjective. (4) now follows as $(\pi \circ f)_* \mathcal{O}_Y(E - \lrcorner \frac{1}{p}F_p \lrcorner + f^*(K_X + \Delta + D)) \subset \pi_* \mathcal{O}_X(K_X + \Delta + D)$ and

$$(\pi \circ f)_* \mathcal{O}_T(E - \lrcorner \frac{1}{p}F_p \lrcorner + f^*(K_X + \Delta + D)) = \pi_* \mathcal{J}_{(\Delta - S)|_S, \|D\|_S}(K_X + \Delta + D).$$

□

4.2. Extending pluricanonical forms.

Theorem 4.11. *Let $\pi: X \rightarrow Y$ be a projective morphism from a smooth quasi-projective variety to an affine variety, $\Delta = \sum \delta_i \Delta_i$ be a divisor with simple normal crossings support and rational coefficients $0 \leq \delta_i \leq 1$ and S an irreducible component of $\lrcorner \Delta \lrcorner$. Let $k > 0$ be an integer such that $D = k(K_X + \Delta)$ is Cartier. If $\mathbf{B}(D)$ contains no log canonical centers of $(X, \lrcorner \Delta \lrcorner)$ and if A is a sufficiently ample Cartier divisor, then*

$$(\star_m) \quad \mathcal{J}_{\|mD\|_S} \subset \mathcal{J}_{(\lrcorner \Delta \lrcorner - S)|_S, \|mD + A\|_S} \quad \text{for all } m > 0, \quad \text{and}$$

$$\pi_* \mathcal{J}_{\|mD\|_S}(mD + A) \subset \text{Im}(\pi_* \mathcal{O}_X(mD + A) \rightarrow \pi_* \mathcal{O}_S(mD + A))$$

for all $m > 0$.

Proof. We proceed by induction on m . Since $\mathcal{J}_{(\lceil \Delta \rceil - S)|_S, \|A\|_S} = \mathcal{O}_X$, $(\star_0)$ is clear. Therefore we must show that $(\star_{m+1})$ holds assuming that $(\star_m)$ holds. There is a unique choice of integral divisors

$$S \leq \Delta^1 \leq \Delta^2 \leq \dots \leq \Delta^k = \Delta^{k+1} = \lceil \Delta \rceil$$

such that $k\Delta = \sum_{i=1}^k \Delta^i$ (cf. (4.23)). We let

$$D_{\leq 0} = 0 \quad \text{and} \quad D_{\leq s} = K_X + \Delta^s + D_{\leq s-1} \quad \text{for } 0 < s \leq k+1.$$

In particular $D = D_{\leq k}$. Since $\mathcal{J}_{\|(m+1)D\|_S} \subset \mathcal{J}_{\|mD\|_S}$ (cf. (1) of (4.10)) and since $\mathcal{J}_{(\Delta^{k+1}-S)|_S, \|mD+D_{\leq k}+A\|_S} = \mathcal{J}_{(\lceil \Delta \rceil - S)|_S, \|(m+1)D+A\|_S}$, to prove $(\star_{m+1})$, it suffices to show that

$$(\sharp_i) \quad \mathcal{J}_{\|mD\|_S} \subset \mathcal{J}_{(\Delta^{i+1}-S)|_S, \|mD+D_{\leq i}+A\|_S} \quad \text{for } 0 \leq i \leq k.$$

We proceed by induction on $0 \leq i \leq k$. $(\sharp_0)$ holds since by $(\star_m)$ and (1) of (4.10), we have

$$\mathcal{J}_{\|mD\|_S} \subset \mathcal{J}_{(\lceil \Delta \rceil - S)|_S, \|mD+A\|_S} \subset \mathcal{J}_{(\Delta^1 - S)|_S, \|mD+A\|_S}.$$

Assume that $(\sharp_{i-1})$ holds for some $0 < i \leq k$. We have that

$$\begin{aligned} \pi_* \mathcal{J}_{\|mD\|_S} (mD + D_{\leq i} + A) &\subset \pi_* \mathcal{J}_{(\Delta^i - S)|_S, \|mD+D_{\leq i-1}+A\|_S} (mD + D_{\leq i} + A) \\ &\subset \text{Im} (\pi_* \mathcal{O}_X (mD + D_{\leq i} + A) \rightarrow \pi_* \mathcal{O}_S (mD + D_{\leq i} + A)) \\ &\subset \pi_* \mathcal{J}_{(\Delta^{i+1} - S)|_S, \|mD+D_{\leq i}+A\|_S} (mD + D_{\leq i} + A). \end{aligned}$$

The first inclusion follows as we are assuming $(\sharp_{i-1})$. The second inclusion follows by (4) of (4.10), since we may assume that $D_{\leq i-1} + A$ is ample and hence that $\mathbf{B}_+(mD + D_{\leq i-1} + A)$ contains no log canonical centers of (X, Δ^i) . The third inclusion follows from (2) of (4.10) since we may assume that $D_{\leq i} + A$ is ample and hence that $\mathbf{B}(mD + D_{\leq i} + A)$ contains no log canonical centers of (X, Δ^{i+1}) .

By (3) of (4.10)

$$\mathcal{J}_{\|mD\|_S} (mD + D_{\leq i} + A) = \mathcal{J}_{\|mD\|_S} (K_S + mD|_S + (\Delta^i - S + D_{\leq i-1} + A)|_S)$$

is generated by global sections since $(\Delta^i - S + D_{\leq i-1} + A)|_S$ is sufficiently ample. Therefore, $(\sharp_i)$ holds. This concludes the proof of $(\star_m)$ for all $m \geq 0$. The remaining claim is clear from what we have shown above. $\square$

Theorem 4.12. *Let $\pi: X \rightarrow Z$ be projective morphism from a smooth quasi-projective variety to an affine variety. Let $\Delta = S + A + B$ be a $\mathbb{Q}$ -divisor such that $S = \lfloor \Delta \rfloor$ is irreducible and smooth, $A \geq 0$ is a general ample $\mathbb{Q}$ -divisor, $B \geq 0$, (X, Δ) is purely log terminal, $(S, \Omega + A|_S)$ is canonical where $\Omega = (A + B)|_S$, and $\mathbf{B}(K_X + \Delta)$ does not contain S . For any sufficiently divisible integer $m > 0$ let*

$$F_m = \text{Fix}(|m(K_X + \Delta)|_S)/m \quad \text{and} \quad F = \lim F_m.$$

Let $\epsilon > 0$ be a rational number such that $\epsilon(K_X + \Delta) + A$ is ample. If Φ is a $\mathbb{Q}$ -divisor on S and $k > 0$ is an integer such that $k\Delta$ and $k\Phi$ are Cartier and

$$\Omega \wedge (1 - \frac{\epsilon}{k})F \leq \Phi \leq \Omega$$

then

$$|k(K_S + \Omega - \Phi)| + k\Phi \subset |k(K_X + \Delta)|_S.$$

Proof. Since A is a general ample $\mathbb{Q}$ -divisor, $\frac{k-1}{k}A$ is a multiple of a general very ample divisor and so $(X, \Delta + \frac{k-1}{k}A)$ is purely log terminal. By assumption $(S, \Omega + \frac{1}{k}A|_S)$ is canonical. Let $l > 0$ be a sufficiently big and divisible integer.

We will show (see (‡) below) that there exists an effective divisor H on X not containing S such that for all integers $m > 0$ divisible by l , we have

$$(†) \quad |m(K_S + \Omega - \Phi)| + m\Phi + (\frac{m}{k}A + H)|_S \subset |m(K_X + \Delta) + \frac{m}{k}A + H|_S.$$

Grant this for the time being. Then, for any $\Sigma \in |k(K_S + \Omega - \Phi)|$, we may choose a divisor $G \in |m(K_X + \Delta) + \frac{m}{k}A + H|$ such that $G|_S = \frac{m}{k}\Sigma + m\Phi + (\frac{m}{k}A + H)|_S$. If we define $\Lambda = \frac{k-1}{m}G + B$, then

$$k(K_X + \Delta) \sim_{\mathbb{Q}} K_X + S + \Lambda + (\frac{1}{k}A - \frac{k-1}{m}H)$$

where $\frac{1}{k}A - \frac{k-1}{m}H$ is ample as m is sufficiently big. By (4.6), we have a surjective homomorphism

$$H^0(X, \mathcal{J}_{S, \Lambda}(k(K_X + \Delta))) \rightarrow H^0(S, \mathcal{J}_{\Lambda|_S}(k(K_X + \Delta))).$$

Since (S, Ω) is canonical, $(S, \Omega + \frac{k-1}{m}H|_S)$ is kawamata log terminal as m is sufficiently big. Therefore $\mathcal{J}_{\Omega + \frac{k-1}{m}H|_S} = \mathcal{O}_S$. Since

$$\Lambda|_S - (\Sigma + k\Phi) = (\frac{k-1}{m}G + B)|_S - (\Sigma + k\Phi) \leq \Omega + \frac{k-1}{m}H|_S,$$

then by (2) of (4.4), we have $\mathcal{I}_{\Sigma + k\Phi} \subset \mathcal{J}_{\Lambda|_S}$ and so

$$\Sigma + k\Phi \in |k(K_X + \Delta)|_S$$

and the Theorem follows.

Let $f: Y \rightarrow X$ be a log resolution of $(X, \Delta + \frac{1}{k}A)$ and of $|l(K_X + \Delta + \frac{1}{k}A)|$. We write

$$K_Y + \Gamma = f^*(K_X + \Delta + \frac{1}{k}A) + E$$

as in (*) of §1.1. Define

$$\Xi = \Gamma - E \wedge \frac{\text{Fix}(l(K_Y + \Gamma))}{l}.$$

We have that $l(K_Y + \Xi)$ is Cartier, $\text{Fix}(l(K_Y + \Xi)) \wedge \Xi = 0$ and $\text{Mob}(l(K_Y + \Xi))$ is free. Since $\text{Fix}(l(K_Y + \Xi)) + \Xi$ has simple normal crossings support, it follows that $\mathbf{B}(K_Y + \Xi)$ contains no log canonical center of $(Y, \lceil \Xi \rceil)$. Let $T = (f^{-1})_*S$. Let $\Gamma_T = (\Gamma - T)|_T$ and $\Xi_T = (\Xi - T)|_T$. Let $m > 0$ be divisible by l . By (4.16)

$$\sigma \in H^0(T, \mathcal{O}_T(m(K_T + \Xi_T))) = H^0(T, \mathcal{J}_{\|m(K_T + \Xi_T)\|}(m(K_T + \Xi_T))).$$

By (4.11), there is an ample divisor H on Y such that if $\tau \in H^0(T, \mathcal{O}_T(H))$, then $\sigma \cdot \tau$ is in the image of the homomorphism

$$H^0(Y, \mathcal{O}_Y(m(K_Y + \Xi) + H)) \rightarrow H^0(T, \mathcal{O}_T(m(K_Y + \Xi) + H)).$$

Therefore

$$(‡) \quad |m(K_T + \Xi_T)| + m(\Gamma_T - \Xi_T) + |H|_T \subset |m(K_Y + \Gamma) + H|_T.$$

We claim that

$$(b) \quad \Omega + \frac{1}{k}A|_S \geq (f|_T)_*\Xi_T \geq \Omega - \Phi + \frac{1}{k}A|_S$$

and so, as $(S, \Omega + \frac{1}{k}A|_S)$ is canonical, we have

$$|m(K_S + \Omega - \Phi)| + m((f|_T)_* \Xi_T - \Omega + \Phi) \subset |m(K_S + (f|_T)_* \Xi_T)| = (f|_T)_* |m(K_T + \Xi_T)|$$

cf. (4.24). Pushing forward the inclusion $(\dagger)$, one sees that

$$|m(K_S + \Omega - \Phi)| + m\Phi + (\frac{m}{k}A + f_*H)|_S \subset |m(K_X + \Delta) + \frac{m}{k}A + f_*H|_S.$$

Equation $(\dagger)$ now follows.

We will now prove the inequality (b) claimed above. We have $\Xi_T \leq \Gamma_T$ and $(f|_T)_* \Gamma_T = \Omega + \frac{1}{k}A|_S$ and so the first inequality follows.

In order to prove the second inequality, let P be any prime divisor on S and let P' be its strict transform on T . Assume that P is contained in the support of Ω , or equivalently, that P' is contained in the support of $(\Gamma - T)|_T$. Then there is a component G of the support of Γ such that

$$\text{mult}_{P'}(\text{Fix}|l(K_Y + \Gamma)|_T) = \text{mult}_G(\text{Fix}|l(K_Y + \Gamma)|)$$

and $\text{mult}_{P'}(\Gamma_T) = \text{mult}_G(\Gamma)$. It follows that

$$\text{mult}_{P'}(\Xi_T) = \text{mult}_G(\Xi) = \text{mult}_G(\Gamma) - \min\{\text{mult}_G(\Gamma), \frac{1}{l}\text{mult}_G(\text{Fix}|l(K_Y + \Gamma)|)\}.$$

Since $\text{mult}_P(\Omega + \frac{1}{k}A|_S) = \text{mult}_{P'}(\Gamma_T)$ and $\Omega \wedge (1 - \frac{\epsilon}{k})F \leq \Phi$, it suffices to show that

$$\text{mult}_{P'}(\text{Fix}|l(K_Y + \Gamma)|_T) \leq l(1 - \frac{\epsilon}{k})\text{mult}_P(F).$$

Since $E|_T$ is exceptional, we have that

$$\text{mult}_{P'}(\text{Fix}|l(K_Y + \Gamma)|_T) = \text{mult}_P(\text{Fix}|l(K_X + \Delta + \frac{1}{k}A)|_S)$$

cf. (4.25). Let $\eta > \epsilon/k$ be a rational number such that $\eta(K_X + \Delta) + \frac{1}{k}A$ is ample. Since l is sufficiently big and divisible and

$$l(K_X + \Delta + \frac{1}{k}A) \cong l(1 - \eta)(K_X + \Delta) + l(\eta(K_X + \Delta) + \frac{1}{k}A),$$

we have that

$$\text{mult}_P(\text{Fix}|l(K_X + \Delta + \frac{1}{k}A)|_S) \leq l(1 - \frac{\epsilon}{k})\text{mult}_P(F)$$

where we have used the fact that if $\text{mult}_P(F) = 0$, then $\text{mult}_P(\text{Fix}|l(K_X + \Delta + \frac{1}{k}A)|_S) = 0$ for all $l \gg 0$ cf. (4.19). $\square$

Exercise 4.13. Notation as in (4.4). Show that if $c < 1$ and $D \in V$ is general, then $\mathcal{J}_{\Delta, c \cdot V} = \mathcal{J}_{\Delta, c \cdot D}$.

Exercise 4.14. Notation as in (4.4). Show that $\mathcal{J}_{\Delta, D} = \mathcal{O}_X$ if and only if $(X, D + \Delta)$ is divisorially log terminal and $\lrcorner D \lrcorner = 0$.

Exercise 4.15. Notation as in (4.4). Show that $\mathcal{J}_{\Delta, c \cdot D} = \mathcal{O}_X$ if $c \ll 1$ (in fact $\mathcal{J}_{c \cdot D} = \mathcal{O}_X$ if $\text{mult}_x(c \cdot D) < 1$ for all $x \in X$ cf. [Lazarsfeld04, 9.5.13]) and that $\mathcal{J}_{\Delta, D} \subset m_x$ if $\text{mult}_x(D) \geq \dim X$.

Exercise 4.16. Let $\pi: X \rightarrow Z$ be a projective morphism from a smooth quasi-projective variety to an affine variety. Let $D \geq 0$ be a divisor on X . Show that $H^0(\mathcal{O}_X(D)) = H^0(\mathcal{J}_{||D||}(D))$.

Exercise 4.17. Let $D \geq 0$ be a Cartier divisor on a smooth variety X and $Z \subset X$ a closed subvariety such that $\lim \text{mult}_Z(|m!D|)/m! = 0$. Show that $Z \not\subset \mathbf{B}_-(D)$.

Exercise 4.18. Use the arguments of [Lazarsfeld04, 9.2.19] to give an alternative proof of the claim in (4.3).

Exercise 4.19. Show that $\mathcal{J}_{\Delta, D} = \mathcal{J}_{\Delta, \{D\}} \otimes \mathcal{O}_X(-\lfloor D \rfloor)$.

Exercise 4.20. Let $f: X \rightarrow Y$ be a birational morphism of normal varieties. If Y is $\mathbb{Q}$ -factorial, show that there exists an effective f -exceptional divisor F such that $-F$ is ample over Y .

Exercise 4.21. Let $f: Y \rightarrow X$ and $\pi: X \rightarrow Z$ be projective morphisms of normal quasi-projective varieties. Show that if F is a coherent sheaf on Y such that $R^j f_* F = 0$ for all $j > 0$, then $R^i(\pi \circ f)_* F \cong R^i \pi_*(f_* F)$ for all $i > 0$.

Exercise 4.22. Let $\pi: X \rightarrow Z$ be a projective morphism from a smooth variety to an affine variety. Let F be a coherent sheaf on X and H be a very ample divisor on X such that $H^i(X, F(mH)) = 0$ for all $i > 0$ and all $m \geq -\dim X$. Show that F is generated by global sections.

Exercise 4.23. Show that if $\Delta = \sum \delta_j \Delta_j$, then the divisors Δ^i in (4.11) are given by $\Delta^i = \sum_{j|\delta_j > (k-s)/k} \Delta_j$.

Exercise 4.24. Let $f: Y \rightarrow X$ be a proper birational morphism of normal varieties. Show that if $\Xi \geq 0$ is a $\mathbb{Q}$ -divisor on Y such that $(X, f_* \Xi)$ is canonical and $m > 0$ is an integer such that $m(K_X + f_* \Xi)$ and $m(K_Y + \Xi)$ are Cartier, then $|m(K_X + f_* \Xi)| = f_* |m(K_Y + \Xi)|$.

Exercise 4.25. Let $f: Y \rightarrow X$ be a proper birational morphism of normal varieties, $S \subset X$ a divisor and $T = (f^{-1})_* S$. Let D be a Cartier divisor on X . Show that $|f^* D|_T = (f|_T)^* |D|_S$. Assume that T and S are normal and that P is a prime divisor on S and P' is its strict transform on T . Show that $\text{mult}_{P'}(\text{Fix } |f^* D|_T) = \text{mult}_P(\text{Fix } |D|_S)$.

5. FINITE GENERATION OF THE RESTRICTED ALGEBRA

5.1. Rationality of the restricted algebra.

Theorem 5.1. Assume Theorem 2.11 _{$n-1$} .

Let $\pi: X \rightarrow Z$ be projective morphism from a smooth quasi-projective variety of dimension n to a normal affine variety. Let $\Delta = S + A + B$ be a $\mathbb{Q}$ -divisor such that $S = \lfloor \Delta \rfloor$ is irreducible and smooth, $A \geq 0$ is a general ample $\mathbb{Q}$ -divisor, $B \geq 0$, (X, Δ) is purely log terminal, $(S, \Omega + A|_S)$ is canonical where $\Omega = (A + B)|_S$, and $\mathbf{B}(K_X + \Delta)$ does not contain S . For any sufficiently divisible integer $m > 0$ let

$$F_m = \text{Fix}(|m(K_X + \Delta)|_S)/m \quad \text{and} \quad F = \lim F_m.$$

Then $\Theta = \Omega - \Omega \wedge F$ is rational and if $k\Delta$ and $k\Theta$ are Cartier, then

$$R_S(k(K_X + \Delta)) \cong R(k(K_S + \Theta)).$$

Proof. Let $V \subset \text{WDiv}_{\mathbb{R}}(S)$ be the sub-vector space generated by the components of Ω and $W \subset V$ be the smallest rational affine subspace containing Θ . By (2.11), there exists a constant $\delta > 0$ such that if $\Theta' \in V$ with $\|\Theta' - \Theta\| \leq \delta$, then any prime divisor contained in $\mathbf{B}(K_S + \Theta)$ is also contained in $\mathbf{B}(K_S + \Theta')$ and there exist a constant $\eta > 0$ and an integer $r > 0$ such that if $\Theta' \in W$ with $\|\Theta' - \Theta\| \leq \eta$ and $k > 0$ is an integer such that $k(K_S + \Theta')/r$ is Cartier, then every component of $\text{Fix}(k(K_S + \Theta'))$ is contained in $\mathbf{B}(K_S + \Theta)$.

Note that if $l(K_X + \Delta)$ is Cartier and $\Theta_l = \Omega - \Omega \wedge F_l$, then $l(K_S + \Theta_l)$ is Cartier and

$$|l(K_X + \Delta)|_S \subset |l(K_S + \Theta_l)| + l(\Omega \wedge F_l).$$

It follows that no component of $\text{Supp}(\Theta_l)$ is contained in $\text{Fix}(l(K_S + \Theta_l))$ and hence in $\mathbf{B}(K_S + \Theta_l)$. For $l \gg 0$, we have that $\Theta_l \in V$, $\|\Theta_l - \Theta\| \leq \delta$ so that no component of $\text{Supp}(\Theta_l)$ is contained in $\mathbf{B}(K_S + \Theta)$. Since $\text{Supp}(\Theta) \subset \text{Supp}(\Theta_l)$, no component of Θ is contained in $\mathbf{B}(K_S + \Theta)$.

Let $0 < \epsilon \ll 1$ be a rational number such that $\epsilon(K_X + \Delta) + A$ is ample. If Θ is not rational, then there exist a component P of Θ with $\text{mult}_P(\Theta) \notin \mathbb{Q}$. Recall that $\Theta = \Omega - \Omega \wedge F$. Notice that if Q is a prime divisor on S such that $\text{mult}_Q(\Omega \wedge F) \notin \mathbb{Q}$, then

$$\text{mult}_Q(\Omega \wedge F) = \text{mult}_Q(F)$$

and if $\text{mult}_Q(\Omega \wedge F) \in \mathbb{Q}$, then $\text{mult}_Q(\Phi) = \text{mult}_Q(\Theta)$ for any $\mathbb{R}$ -divisor $\Phi \in W$.

By Diophantine approximation, there exist an integer $k > 0$, an effective $\mathbb{Q}$ -divisor $\Phi \in W$ such that $k\Phi/r$ and $k\Delta/r$ are Cartier, $\text{mult}_P(\Phi) < \text{mult}_P(\Omega \wedge F)$ and $\|\Phi - (\Omega \wedge F)\| \leq \min\{\eta, \gamma, \epsilon f/k\}$ where f is the smallest non-zero coefficient of F and γ is the smallest non-zero coefficient of $\Omega - \Omega \wedge F$. Then

$$\Omega \wedge (1 - \frac{\epsilon}{k})F \leq \Phi \leq \Omega.$$

By (4.12), we have that

$$(b) \quad |k(K_S + \Omega - \Phi)| + k\Phi \subset |k(K_X + \Delta)|_S.$$

Since

$$\text{mult}_P(\text{Fix}(|k(K_X + \Delta)|_S)) \geq \text{mult}_P(kF) = \text{mult}_P(k(\Omega \wedge F)) > \text{mult}_P(k\Phi),$$

it follows from (b) that P is contained in the support of $\text{Fix}(k(K_S + \Omega - \Phi))$. Since $\|(\Omega - \Phi) - \Theta\| \leq \eta$ and $k(K_S + \Omega - \Phi)/r$ is Cartier, P is contained in $\mathbf{B}(K_S + \Theta)$. This is a contradiction and therefore Θ is rational. The isomorphism $R_S(k(K_X + \Delta)) \cong R(k(K_S + \Theta))$ now follows from (4.12). $\square$

5.2. Proof of (3.9).

Proof of (3.9). We may assume that Z is affine cf. (3.2). By (3.6), it suffices to show that the restricted algebra $R_S(K_X + \Delta)$ is finitely generated. By (3.7), this is equivalent to showing that the restricted algebra $R_S(k(K_X + \Delta))$ is finitely generated for some integer $k > 0$. It follows that we may replace Δ by a $\mathbb{Q}$ -linearly equivalent divisor Δ' . By (3.11), S is mobile. Since f is birational and Z is affine, $\Delta - S$ is big so that $\Delta - S \sim_{\mathbb{Q}} A' + B'$ where A' is ample, $B' \geq 0$ and $\text{Supp}(B')$ does not contain S . Let $\epsilon > 0$ be a sufficiently small rational number. We may replace Δ by

$$\Delta' = S + A + B \sim_{\mathbb{Q}} \Delta$$

where $A \sim_{\mathbb{Q}} \epsilon A'$ is a general ample $\mathbb{Q}$ -divisor and $B = (1 - \epsilon)(\Delta - S) + \epsilon B'$. Let $f: Y \rightarrow X$ be a log resolution of (X, Δ') and write

$$K_Y + \Gamma' = f^*(K_X + \Delta') + E'$$

as in (*) of Notation. Let $T = (f^{-1})_* S$. We may assume that (Y, Γ') is purely log terminal and that $(T, (\Gamma' - T)|_T)$ is terminal. Let $F \geq 0$ be an effective exceptional $\mathbb{Q}$ -divisor such that $f^*A - F$ is ample, $(Y, \Gamma' + F)$ is purely log terminal and $(T, (\Gamma' - T + F)|_T)$ is terminal. We set $\Gamma = \Gamma' - f^*A + F + H$ where $H \sim_{\mathbb{Q}} f^*A - F$ is a general ample $\mathbb{Q}$ -divisor. Then (Y, Γ) is purely log terminal and we may assume

that $(T, (\Gamma - T + H)|_T)$ is terminal. Since $\mathbf{B}(K_X + \Delta)$ does not contain S , one sees that $\mathbf{B}(K_Y + \Gamma)$ does not contain T . By (5.1), it then follows that there is a $\mathbb{Q}$ -divisor Θ on T such that $R_T(k(K_Y + \Gamma)) \cong R(k(K_T + \Theta))$ for some $k > 0$ sufficiently divisible. The Theorem now follows since for $k > 0$ sufficiently divisible,

$$R_T(k(K_Y + \Gamma)) \cong R_S(k(K_X + \Delta)).$$

□

6. LOG TERMINAL MODELS

In this section we will prove the following result:

Theorem 6.1. *Theorems A, B, C, D, E in dimension $n - 1$ and Theorem A in dimension n imply Theorem B in dimension n .*

Proof. See (6.2). □

6.1. Special termination.

Lemma 6.2. *Assume Theorems A, B, C, D, E in dimension $n - 1$. Let $\pi : X \rightarrow U$ be a projective morphism of normal quasi-projective varieties. Let $(X, \Delta + C = S + A + B + C)$ be a $\mathbb{Q}$ -factorial divisorially log terminal pair with $S = \lfloor \Delta \rfloor$, such that $K_X + \Delta + C$ is nef over U and $\mathbf{B}_+(A/U)$ contains no log canonical centers of $(X, \Delta + C)$. Let $\phi_i : X_i \dashrightarrow X_{i+1}$ be a sequence of flips and divisorial contractions over U for the $K_X + \Delta$ minimal model program over U with scaling of C .*

Then there exists an integer $k > 0$ such that for all $i \geq k$, ϕ_i is an isomorphism on a neighbourhood of S .

Proof. Suppose that this is not the case, then we may assume that infinitely often ϕ_i is a flip and the flipping locus intersects a component S' of S .

Step 1. We may assume that S is irreducible and A is a general ample $\mathbb{Q}$ -divisor over U .

Since $\mathbf{B}_+(A/U)$ contains no log canonical centers of $(X, \Delta + C)$, we have $A \sim_{\mathbb{R}, U} A' + B'$ where A' is a $\mathbb{Q}$ -divisor ample over U and $\text{Supp}(B')$ contains no log canonical centers of $(X, \Delta + C)$. For any rational number $0 < \epsilon \ll 1$, $A' + \epsilon(S - S')$ is an ample $\mathbb{Q}$ -divisor. Let $A'' \sim_{\mathbb{R}, U} A' + \epsilon(S - S')$ be a general ample $\mathbb{Q}$ -divisor over U . For any rational number $0 < \delta \ll 1$, we have $S + A + B \sim_{\mathbb{R}, U} S' + \delta A'' + (1 - \epsilon\delta)(S - S') + (1 - \delta)A + \delta B'$ where $\delta A''$ is a general ample $\mathbb{Q}$ -divisor over U and $(X, S' + \delta A'' + (1 - \epsilon\delta)(S - S') + (1 - \delta)A + \delta B' + C)$ is divisorially log terminal. Therefore, we may assume that S is irreducible and A is a general ample $\mathbb{Q}$ -divisor over U .

Let S_i, A_i, B_i and C_i be the strict transforms of S, A, B and C on X_i .

Step 2. We may assume that for $i \gg 0$, the birational map $S_i \dashrightarrow S_{i+1}$ is an isomorphism in codimension 1.

Let $d = \text{disc.}(X, \Delta) \in (-1, 0]$, $\Delta = \sum_{i \in I} \delta_i \Delta_i$ and

$$\mathcal{B} = \{1 - \frac{1}{m} + \sum_{i \in I} \frac{r_i \delta_i}{m} \in [0, -d] \mid m \in \mathbb{Z}_{>0}, r_i \in \mathbb{Z}_{\geq 0}\}.$$

Then $\mathcal{B}$ is a finite set. We define (cf. [Fujino07])

$$d_{\mathcal{B}}(S, \Delta_S) := \sum_{\beta \in \mathcal{B}} \#\{E \mid a(E, S, \Delta_S) < -\beta\} < \infty.$$

Note that $d_{\mathcal{B}}(S_i, \Delta_{S_i}) \leq d_{\mathcal{B}}(S_{i+1}, \Delta_{S_{i+1}})$. Therefore, we may assume that $d_{\mathcal{B}}(S_i, \Delta_{S_i})$ is constant for $i \gg 0$.

Suppose that $P \subset S_{i+1}$ is a divisor such that $(\phi_i^{-1})_* P = 0$. Since $-a(P, S_{i+1}, \Delta_{S_{i+1}}) \in \mathcal{B}$ cf. (1.5), then the above inequality is strict. Therefore $S_i \dashrightarrow S_{i+1}$ extracts no divisors for all $i \gg 0$. If $S_i \dashrightarrow S_{i+1}$ contracts a divisor, then $\rho(S_i) > \rho(S_{i+1})$. This can happen at most finitely many times and so Step 2 follows.

Step 3. There exist integers $j > i$ such that $S_i \dashrightarrow S_{i+1}$ is not an isomorphism but $S_i \dashrightarrow S_j$ is an isomorphism.

By assumption, there is a non-increasing sequence of numbers λ_i such that $K_X + \Delta_i + \lambda_i C_i$ is nef over U . Since $K_{S_i} + \Delta_{S_i} + \lambda_i C_i|_{S_i}$ is nef over U and $S_i \dashrightarrow S_{i+1}$ is an isomorphism in codimension 1, by (2.16), there are finitely many rational maps $S_i \dashrightarrow S_j$ (modulo isomorphism). Let $X_i \rightarrow Z_i$ be the flipping contraction. Let T_i be the normalization of the image of S_i in Z_i . Suppose that $S_i \rightarrow T_i$ is an isomorphism and the flipping locus intersects S_i , then $\Sigma_i \cdot S_i > 0$ for any flipping curve $\Sigma_i \subset X_i$ and so $\Sigma_{i+1} \cdot S_{i+1} < 0$ for any flipped curve $\Sigma_{i+1} \subset X_{i+1}$ and so $S_{i+1} \rightarrow T_i$ is not an isomorphism. Therefore, infinitely often $S_i \dashrightarrow S_{i+1}$ is not an isomorphism. Step 3 now follows.

Step 4. We find the required contradiction.

Let E be a divisor with center contained in the flipping locus of $X_i \dashrightarrow X_{i+1}$, then $a(E, S_i, \Delta_{S_i}) < a(E, S_{i+1}, \Delta_{S_{i+1}})$. Since $a(E, S_{i+1}, \Delta_{S_{i+1}}) \leq a(E, S_j, \Delta_{S_j})$, we have the required contradiction. $\square$

Lemma 6.3. *Assume Theorems A, B, C, D, E in dimension $n-1$ and Theorem A in dimension n . Let $\pi : X \rightarrow U$ be a projective morphism of normal quasi-projective varieties. Let $(X, \Delta + C = S + A + B + C)$ be a $\mathbb{Q}$ -factorial divisorially log terminal pair with $S = \lfloor \Delta \rfloor$, such that $K_X + \Delta + C$ is nef over U and $\mathbf{B}_+(A/U)$ contains no log canonical centers of $(X, \Delta + C)$. Suppose that there is an effective $\mathbb{R}$ -divisor D such that $\text{Supp}(D) \subset S$ and a non-negative number α such that*

$$(\sharp) \quad K_X + \Delta \sim_{\mathbb{R}, U} D + \alpha C.$$

Then there exists a log terminal model $\phi : X \dashrightarrow Y$ for $K_X + \Delta$ over U such that $\mathbf{B}_+(\phi_ A/U)$ contains no log canonical centers of $(Y, \phi_* \Delta)$.*

Proof. We run the $K_X + \Delta$ minimal model program over U with scaling of C . If R is a $K_X + \Delta$ negative extremal ray such that $(K_X + \Delta + \lambda C) \cdot R = 0$, then $D \cdot R < 0$ and so R induces a flipping or divisorial contraction.

Notice that if $f_i : X_i \rightarrow Z_i$ is a flipping contraction, as $D_i \cdot R_i < 0$, there is a component S' of S such that $S' \cdot R_i < 0$. But then f_i is a pl-flipping contraction for some divisorially log terminal pair $K_X + S' + (1-\epsilon)(S-S') + A' + B'$ where $0 < \epsilon \ll 1$ and A', B' are appropriate $\mathbb{Q}$ -divisors with $\|A - A'\| \ll 1$ and $\|B - B'\| \ll 1$. Therefore, by Theorem A_n, the flip $X_i \dashrightarrow X_{i+1}$ exists.

Note that condition $(\sharp)$ is preserved by flips and divisorial contractions. Since $S' \cdot R_i < 0$, $X_i \rightarrow Z_i$ is not an isomorphism on a neighborhood of S . By (6.3), the sequence of flips and divisorial contractions terminates. $\square$

6.2. Existence of log terminal models.

Proof of (6.1). Step 1. We may assume that X is smooth, $\text{Supp}(\Delta + D)$ is simple normal crossings, $\mathbf{B}_+(A/U)$ contains no log canonical centers for $(X, \Delta = A + B)$ and if G is a component of $\text{Supp}(D)$ then either G is contained in $\mathbf{B}(D/U)$ or $\text{Fix}(G) = 0$.

Proof: Exercise. (Note that if D is Cartier and $U = \text{Spec}(\mathbb{C})$, then for $m \gg 0$, $|mD| = \text{Fix}(mD) + |M|$ where $\text{Supp}(\text{Fix}(mD)) = \mathbf{B}(D)$ and $\mathbf{Fix}(|M|) = 0$. The case when D is $\mathbb{R}$ -Cartier may be found in [BCHM06]).

Step 2. There exists a rational map $\phi: X \dashrightarrow Y$ over U such that

- (1) ϕ extracts no divisors;
- (2) If P is a prime divisor contracted by ϕ , then $P \subset \text{Supp}(D)$;
- (3) Y is $\mathbb{Q}$ -factorial and $Y \rightarrow U$ is projective;
- (4) $K_Y + \phi_*\Delta$ is divisorially log terminal and nef over U ; and
- (5) No log canonical center of $K_Y + \phi_*\Delta$ is contained in $\mathbf{B}_+(\phi_*A/U)$.

We write $D = D_1 + D_2$ where the components of D_1 are contained in $\text{Supp}(\lfloor \Delta \rfloor)$ and the components of D_2 are not contained in $\text{Supp}(\lfloor \Delta \rfloor)$. We proceed by induction on the number of components of D_2 .

If $D_2 = 0$, then pick H a general element of a sufficiently ample divisor. We may assume that $K_X + \Delta + H$ is divisorially log terminal and nef over U . We may then run the minimal model program over U for $K_X + \Delta$ with scaling of H . By (6.3), there exists a log terminal model over U , $\phi: X \dashrightarrow Y$. It is easy to see that ϕ satisfies properties (1-5) above.

If $D_2 \neq 0$, let

$$\lambda = \sup\{t \geq 0 \mid (X, \Delta + tD_2) \text{ is DLT}\}.$$

Then $(X, \Delta' = \Delta + \lambda D_2)$ is divisorially log terminal and

$$K_X + \Delta' \sim_{\mathbb{R}, U} D' = D + \lambda D_2.$$

Note that $\text{Supp}(D') = \text{Supp}(D)$ and the number of components of D' not contained in $\lfloor \Delta' \rfloor$ is less than the number of components of D not contained in $\lfloor \Delta \rfloor$. By induction, there is a rational map $\phi': X \dashrightarrow Y'$ over U such Δ' and D' satisfy properties (1-5) above. Since $\lambda > 0$, $K_{Y'} + \phi'_*(\Delta + \lambda D_2)$ is divisorially log terminal and nef over U , since $\mathbf{B}_+(\phi'_*A/U)$ contains no log canonical center of $(Y', \phi'_*(\Delta + \lambda D_2))$ and since

$$K_{Y'} + \phi'_*\Delta \sim_{\mathbb{R}, U} \phi'_*D_1 + \phi'_*D_2$$

then by (6.3), there exists a log terminal model $f: Y' \dashrightarrow Y$ for $K_{Y'} + \phi'_*\Delta$ over U . Let $\phi = f \circ \phi': X \dashrightarrow Y$. Then ϕ satisfies properties (1-5) above.

Step 3. $\phi: X \dashrightarrow Y$ is a log terminal model for (X, Δ) over U .

Let $p: W \rightarrow X$ and $q: W \rightarrow Y$ be proper birational maps of normal varieties such that $\phi = q \circ p^{-1}$. Then

$$p^*(K_X + \Delta) + E = q^*(K_Y + \phi_*\Delta) + F$$

where E and F are effective q -exceptional $\mathbb{R}$ -divisors with no common components.

We have that $K_Y + \phi_*\Delta$ is semiample over U cf. (1.14). Therefore, $\mathbf{B}(q^*(K_Y + \phi_*\Delta) + F/U) = \text{Supp}(F)$. On the other hand $\text{Supp}(E) \subset \mathbf{B}(p^*(K_X + \Delta) + E/U)$. Therefore, $E = 0$ and $p_*(E)$ contains all ϕ -exceptional divisors. It follows that $a(P, X, \Delta) < a(P, Y, \phi_*\Delta)$ for all ϕ -exceptional divisors $P \subset X$ and so Y is a log terminal model for (X, Δ) over U . □

7. NON-VANISHING

Recall the following results from [Nakayama04].

7.1. Nakayama-Zariski Decomposition.

Definition 7.1. Let X be a smooth variety, D be a big divisor on X and Q be a prime divisor on X . Then

$$\sigma_Q(D) = \inf \{ \text{mult}_Q(D') \mid D \sim_{\mathbb{R}} D' \geq 0 \}, \quad \text{and}$$

$$N_\sigma(D) = \sum \sigma_Q(D) \cdot Q.$$

Remark 7.2. By [Nakayama04, §III.1], it is known that $\sigma_Q(D) = \sigma_Q(D')$ if $D \equiv D'$ and that $\sigma_Q(\dots)$ is a continuous function on the cone of big divisors.

Definition 7.3. Let X be a smooth variety, D be a pseudo-effective divisor on X and Q be a prime divisor on X . Then

$$\sigma_Q(D) = \lim_{\epsilon \rightarrow 0^+} \sigma_Q(D + \epsilon A).$$

Remark 7.4. By [Nakayama04, §III.1], $\sigma_Q(D)$ is independent of the choice of A and there are only finitely many prime divisors Q such that $\sigma_Q(D) > 0$. Therefore, $N_\sigma(D) = \sum \sigma_Q(D) \cdot Q$ is a well defined $\mathbb{R}$ -divisor. If $D \equiv D'$ then $N_\sigma(D) = N_\sigma(D')$. We have that $D - N_\sigma(D)$ is pseudo-effective and $N_\sigma(D - N_\sigma(D)) = 0$. Finally, if Q is an effective $\mathbb{R}$ -divisor whose support is contained in $\text{Supp}(N_\sigma(D))$, then $N_\sigma(D + Q) = N_\sigma(D) + Q$.

Lemma 7.5. *Let X be a smooth variety, D a pseudo-effective $\mathbb{R}$ -divisor, G a $\mathbb{Q}$ -divisor such that $A = G - D$ is an ample $\mathbb{R}$ -divisor and Q a prime divisor. If $\sigma_Q(D) = 0$, then Q is not contained in $\mathbf{B}(G)$. In particular Q is not contained in $\mathbf{B}_-(D)$.*

Proof. For any $l > 0$ sufficiently divisible we have that $\frac{l}{2}A - (\dim X + 1)H - K_X$ is ample where H is very ample and $l(D + A/2) \sim_{\mathbb{R}} \Delta$ where $\text{mult}_P(\Delta) < 1$. Therefore $\mathcal{J}_\Delta(lG)$ is generated by global sections cf. (4.22). The lemma now follows as P is not contained in the co-support of $\mathcal{J}_\Delta$ cf. (4.15). $\square$

One of the key results that we will need is the following:

Theorem 7.6. *Let X be a smooth variety, and D be a pseudo-effective $\mathbb{R}$ -divisor on X . Then the following are equivalent:*

- (1) $D \equiv N_\sigma(D)$;
- (2) *For any ample divisor A , the function $h^0(X, \mathcal{O}_X(\lfloor tD \rfloor + A))$ is bounded;*
- (3) *For any ample divisor A , $\lim_{t \rightarrow \infty} \frac{1}{t} h^0(X, \mathcal{O}_X(\lfloor tD \rfloor + A)) = 0$.*

Proof. We follow [Nakayama04, V.1.12]. We begin by showing that (1) implies (2). Since $P = D - N_\sigma(D) \equiv 0$, one sees that there exists an ample divisor A' such that $|A' - A + \lfloor tP \rfloor| \neq \emptyset$ for any integer $t \geq 0$. Therefore $h^0(\mathcal{O}_X(\lfloor tD \rfloor + A)) \leq h^0(\mathcal{O}_X(\lfloor tN_\sigma(D) \rfloor + A'))$. For any divisor Q in the support of $N_\sigma(D)$, and any $k \gg 0$, we have that $\sigma_Q(kN_\sigma(D) + A) > 0$. By (7.4), it follows that for $t > k$, we have $\sigma_\Gamma(tN_\sigma(D) + A') > (t - k)\sigma_\Gamma(N_\sigma(D))$. In particular $h^0(\mathcal{O}_X(\lfloor tN_\sigma(D) \rfloor + A')) = h^0(\mathcal{O}_X(\lfloor kN_\sigma(D) \rfloor + A'))$.

Clearly (2) implies (3).

We will now see that (3) implies (1). Replacing D by $D - N_\sigma(D)$, we may assume that $N_\sigma(D) = 0$. Let A be a very ample divisor and C be a curve given by the intersection of $n - 1 = \dim X - 1$ general elements of $|A|$. We fix an ample divisor A' such that $\frac{1}{2}A' - K_X - \{mD\} - (n - 1)A$ is ample for all $m \geq 0$. Let $H_i \in |A|$ be general elements vanishing along C , then $\mathcal{J}(\frac{n-1}{n}(H_1 + \dots + H_n)) = \mathcal{I}_C$.

Since $\mathbf{B}_-(D)$ is the union of countably many subvarieties of codimension at least 2, $\mathbf{B}_-(D) \cap C = \emptyset$. Therefore, there exists a $\mathbb{R}$ -divisor $\Delta \sim mD + \frac{1}{2}A'$ such that $\mathcal{J}(\Delta + \frac{n-1}{n}(H_1 + \dots + H_n)) = \mathcal{I}_C$. Therefore, we have that $H^1(\mathcal{I}_C(A')) = 0$ and so $H^0(\mathcal{O}_X(\lfloor mD \rfloor + A')) \rightarrow H^0(\mathcal{O}_C(\lfloor mD \rfloor + A'))$ is surjective. But then $\lim_{\frac{1}{m}} h^0(\mathcal{O}_C(\lfloor mD \rfloor + A')) = 0$ so that $D \cdot C = 0$. Since D is pseudo-effective, $D \equiv 0$. $\square$

We will also need the following results concerning Nakayama-Zariski decomposition of the restricted linear series $|D|_S$.

Definition 7.7. Let S be a smooth divisor on a smooth projective variety X and let D be a big $\mathbb{R}$ -divisor on X such that S is not contained in $\mathbf{B}_+(D)$, then we define

$$\sigma_P(\|D\|_S) = \inf\{\text{mult}_P(D'|_S) \mid D \sim_{\mathbb{R}} D' \geq 0, S \not\subset \text{Supp}(D')\}, \quad \text{and}$$

$$N_\sigma(\|D\|_S) = \sum \sigma_P(\|D\|_S) \cdot P.$$

We have the following:

Lemma 7.8. *Let S be a smooth divisor on a smooth projective variety X .*

- (1) *If A is an ample $\mathbb{R}$ -divisor and D is a big $\mathbb{R}$ -divisor such that $S \not\subset \mathbf{B}(D)$, then $\sigma_P(\|D + A\|_S) \leq \sigma_P(\|D\|_S)$. In particular $N_\sigma(\|A\|_S) = 0$.*
- (2) *If D is a big $\mathbb{R}$ -divisor such that $S \not\subset \mathbf{B}_+(D)$ and P is a prime divisor on S , then $\lim_{\epsilon \rightarrow 0^+} \sigma_P(\|D + \epsilon A\|_S) = \sigma_P(\|D\|_S)$.*
- (3) *If D is a pseudo-effective $\mathbb{R}$ -divisor such that $S \not\subset \mathbf{B}_-(D)$ and P is a prime divisor on S , then $\lim_{\epsilon \rightarrow 0^+} \sigma_P(\|D + \epsilon A\|_S) = \sigma_P(\|D\|_S)$ exists and is independent of A .*

Proof. The first statement is clear as $A \sim_{\mathbb{R}} \sum r_i A_i$ and $r_i \in \mathbb{R}_{>0}$ and $k_i A_i$ are very ample for any $k_i > 0$ sufficiently divisible.

For the second statement, note that clearly $\sigma_P(\|D + \epsilon A\|_S) \leq \sigma_P(\|D\|_S)$ for any prime divisor P on S . Let C be an effective $\mathbb{Q}$ -divisor whose support does not contain S and $\delta > 0$ be a rational number such that $D \sim_{\mathbb{Q}} \delta A + C$. For $\epsilon \geq 0$, we have

$$(1 + \epsilon)\sigma_P(\|D\|_S) \leq \sigma_P(\|D + \epsilon \delta A\|_S) + \epsilon \text{mult}_P(C).$$

The assertion now follows by taking the limit as $\epsilon \rightarrow 0^+$.

For the last statement, notice that the limits $\lim_{\epsilon \rightarrow 0^+} \sigma_P(\|D + \epsilon A\|_S)$ exist since $\sigma_P(\|D + \epsilon A\|_S)$ is monotonic increasing and bounded. Here, the boundedness follows easily as

$$(D + \epsilon A) \cdot S \cdot A^{\dim X - 2} \geq \sum \sigma_P(\|D + \epsilon A\|_S) \cdot P \cdot (A|_S)^{\dim X - 2}.$$

Since for any ample $\mathbb{R}$ -divisors A and A' , there is a number $\delta > 0$ such that $A - \delta A'$ is ample, it follows that

$$\sigma_P(\|D + \epsilon A\|_S) \leq \sigma_P(\|D + \epsilon \delta A'\|_S)$$

and so that

$$\lim_{\epsilon \rightarrow 0^+} \sigma_P(\|D + \epsilon A\|_S) \leq \lim_{\epsilon \rightarrow 0^+} \sigma_P(\|D + \epsilon A'\|_S).$$

By symmetry, we have the required equality. $\square$

Definition 7.9. If D is a pseudo-effective $\mathbb{R}$ -divisor such that $S \not\subset \mathbf{B}_-(D)$, then we define

$$N_\sigma(\|D\|_S) = \lim_{\epsilon \rightarrow 0^+} N_\sigma(\|D + \epsilon A\|_S).$$

Remark 7.10. Note that $N_\sigma(\|D\|_S) = \sum_{P \subset X} \sigma_P(\|D\|_S) \cdot P$ and that the set of prime divisors $P \subset S$ such that $\sigma_P(\|D\|_S) > 0$ is countable.

Remark 7.11. If $S \not\subset \mathbf{B}_+(D)$ and D' is a $\mathbb{Q}$ -divisor such that $D' \equiv D$, then $S \not\subset \mathbf{B}_+(D')$ and $N_\sigma(\|D\|_S) = N_\sigma(\|D'\|_S)$.

This follows from the fact that if A is an ample $\mathbb{R}$ -divisor, then $A + D - D'$ is also ample. Therefore

$$N_\sigma(\|D\|_S) = \lim_{\epsilon \rightarrow 0^+} N_\sigma(\|D + \epsilon A\|_S) = \lim_{\epsilon \rightarrow 0^+} N_\sigma(\|D' + \epsilon A'\|_S) = N_\sigma(\|D'\|_S)$$

where $A' = A + D - D'$.

Remark 7.12. Let $\mathcal{C} \subset N_{\mathbb{R}}^1(X)$ be the cone of $\mathbb{R}$ -divisors on X such that S is not contained in $\mathbf{B}_+(D)$ and $\mathcal{C}' \subset N_{\mathbb{R}}^1(X)$ be the cone of $\mathbb{R}$ -divisor on X such that S is not contained in $\mathbf{B}_-(D)$, then for any prime divisor P on S , the functions $\sigma_P(\|\cdot\|_S)$ are continuous on $\mathcal{C}$ and lower semi-continuous on $\mathcal{C}'$.

Lemma 7.13. *Let D be a pseudo-effective $\mathbb{R}$ -divisor such that S is not contained in $N_\sigma(D)$.*

Then S is not contained in $\mathbf{B}_-(D)$ and $D|_S - N$ is pseudo-effective for any divisor $N = \sum_{i=1}^k n_i P_i \leq N_\sigma(\|D\|_S)$.

Proof. Let G be any $\mathbb{Q}$ -divisor such that $G - D$ is ample. For $m > 0$ sufficiently divisible we may write $m(G - D) = K_X + (\dim X + 1)H + K$ where K is ample and H is very ample. Since $N_\sigma(D)$ does not contain S , there is a $\mathbb{R}$ -divisor $\Delta \sim_{\mathbb{R}} mD + K$ such that $\text{mult}_S(\Delta) < 1$. But then $\mathcal{J}_\Delta$ does not vanish along S cf. (4.15). Since $mG - \Delta \sim_{\mathbb{R}} K_X + (\dim X + 1)H$, then $\mathcal{O}_X(mG) \otimes \mathcal{J}_\Delta$ is generated by global sections cf. (4.22) and so S is not contained in $\mathbf{B}(G)$. It follows that S is not contained in $\mathbf{B}_-(D)$.

Since $|G|_S \neq \emptyset$, we have that

$$G|_S - \sum_{i=1}^k \sigma_{P_i}(\|G\|_S) \cdot P_i \geq G|_S - N_\sigma(\|G\|_S) \geq 0.$$

Therefore $D|_S - \sum_{i=1}^k \sigma_{P_i}(\|D\|_S) \cdot P_i$ is a limit of effective divisors and hence is pseudo-effective. Since $N \leq \sum_{i=1}^k \sigma_{P_i}(\|D\|_S) \cdot P_i$, the lemma follows. $\square$

Lemma 7.14. *Let D be a pseudo-effective $\mathbb{R}$ -divisor such that S is not contained in $N_\sigma(D)$ and A an ample $\mathbb{R}$ -divisor such that $D + A$ is $\mathbb{Q}$ -Cartier. Let P be a prime divisor on S . If $\sigma_P(\|D\|_S) > 0$, then there is a positive integer l such that $\text{mult}_P \text{Fix}(l(D + A))|_S < \sigma_P(\|D\|_S)$ and if $\sigma_P(\|D\|_S) = 0$, then there is a positive integer l such that $\text{mult}_P \text{Fix}(l(D + A))|_S = 0$.*

Proof. Suppose that $\sigma_P(\|D\|_S) = 0$. Fix $l > 0$ such that $(\dim X + 1)H = \frac{l}{2}A - (K_X + S)$ is ample and let $\Delta \sim_{\mathbb{R}} D + A/2$ be an $\mathbb{R}$ -divisor whose support does not contain S such that $\text{mult}_P(\Delta|_S) < \frac{1}{l}$. Since $l(D + A) \sim_{\mathbb{R}} K_X + S + (\dim X + 1)H + l\Delta$, $\mathcal{J}_{l\Delta|_S}(l(D + A))$ is globally generated cf. (4.22) and its sections lift to $H^0(X, \mathcal{O}_X(l(D + A)))$ cf. (4.6). Since $\text{mult}_P(l\Delta|_S) < 1$, $\mathcal{J}_{l\Delta|_S}$ does not vanish along P and so $\text{mult}_P \text{Fix}(l(D + A))|_S = 0$. The remaining assertion is similar. $\square$

7.2. Non-vanishing.

Theorem 7.15. *Theorems A, B, C, D, E in dimension $n - 1$ and Theorems A, B in dimension n imply Theorem C in dimension n .*

Proof. **Step 1.** We may assume that X is projective.

Let F be a general fiber of π , then $K_F + \Delta_F = (K_X + \Delta)|_F$ is kawamata log terminal and Δ is big. Suppose that $K_F + \Delta_F \sim_{\mathbb{R}} D_F \geq 0$. Since F is a general fiber, there is an open subset U^0 of U and an effective divisor D^0 on $X^0 = \pi^{-1}(U^0)$ such that $(K_X + \Delta)|_{X^0} - D^0 \sim_{\mathbb{R}} 0$. Therefore, there is a divisor D on X such that $K_X + \Delta \sim_{\mathbb{R}} D$ and $D|_{X^0} = D^0 \geq 0$. Therefore, there is an effective divisor H on U such that $K_X + \Delta \sim_{\mathbb{R}, U} D + \pi^*H \geq 0$.

Step 2. We may assume that X is smooth, $\Delta = A + B$ where A is a general ample $\mathbb{Q}$ -divisor and $\text{Supp}(\Delta)$ has simple normal crossings.

Since Δ is big, we may assume that $\Delta = A + B$ where A is a general ample $\mathbb{Q}$ -divisor. Let $f : Y \rightarrow X$ be a log resolution of (X, Δ) such that there is an effective and exceptional $\mathbb{Q}$ -divisor F such that $f^*A - F$ is ample. Let $C \sim_{\mathbb{Q}} f^*A - F$ be a general ample $\mathbb{Q}$ -divisor. We may write

$$K_Y + \Gamma = f^*(K_X + \Delta) + E$$

as in (*) of §1.1. We may assume that $(Y, \Gamma' = \Gamma - f^*A + F + C)$ is kawamata log terminal. Note that if $K_Y + \Gamma' \sim_{\mathbb{R}, U} D \geq 0$, then $K_X + \Delta \sim_{\mathbb{R}, U} f_*D \geq 0$.

Step 3. We may assume that there is an integer $k > 0$ such that $h^0(\mathcal{O}_X(\lfloor mk(K_X + \Delta) \rfloor + kA))$ is an unbounded function of m .

If this is not the case, then by (7.6), $K_X + \Delta \equiv N_\sigma(K_X + \Delta)$. Therefore $A'' = A + N_\sigma(K_X + \Delta) - (K_X + \Delta)$ is ample. Let $A' \sim_{\mathbb{R}} A''$ be an effective divisor such that $(X, \Delta' = A' + \Delta - A)$ is kawamata log terminal. We have that $K_X + \Delta' \sim_{\mathbb{R}} N_\sigma(K_X + \Delta) \geq 0$. By (2.8), there is a log terminal model $\phi : X \dashrightarrow Y$ for (X, Δ') . Since $K_X + \Delta \equiv K_X + \Delta'$ this is also a log terminal model for (X, Δ) . By (1.6), $K_Y + \phi_*\Delta \sim_{\mathbb{R}} D' \geq 0$ and hence there is an effective $\mathbb{R}$ divisor D on X such that $K_X + \Delta \sim_{\mathbb{R}} D$.

Step 4. We may assume that

- (1) (X, Δ) is purely log terminal and $\Delta = S + A + B$ where A is a general ample $\mathbb{Q}$ -divisor and $S = \lfloor \Delta \rfloor$;
- (2) $(S, \Delta_S + A|_S)$ is terminal where $(K_X + \Delta)|_S = K_S + \Delta_S$; and
- (3) $\text{Supp}(\Delta)$ and $\text{Supp}(N_\sigma)$ have no common components.

Let $x \in X$ be a general point. By Step 3, we may assume that there is a divisor

$$H' \in \lfloor \lfloor mk(K_X + \Delta) \rfloor + kA \rfloor \quad \text{s. t.} \quad \text{mult}_x(H') > kn.$$

It follows that there is an effective $\mathbb{R}$ -divisor $H \sim_{\mathbb{R}} m(K_X + \Delta) + A$ with $\text{mult}_x(H) > n$ and hence such that $(X, \Delta + H)$ is not log canonical at x . For any $t \geq 0$, we may write

$$(t+1)(K_X + \Delta) = K_X + \frac{m-t}{t}A + B + t(K_X + \Delta + \frac{1}{m}A) \sim_{\mathbb{R}} K_X + \frac{m-t}{t}A + B + \frac{t}{m}H = K_X + \Delta_t.$$

For $0 < \epsilon \ll 1$, we let $A' = \frac{\epsilon}{m}A$ and $u = m - \epsilon$. It follows that

- (1) (X, Δ_0) is kawamata log terminal;
- (2) $\Delta_t \geq A'$ for any $t \in [0, u]$;

(3) x is contained in a log canonical center of (X, Δ_u) .

Let $f : Y \rightarrow X$ be a log resolution of $(X, \Delta + H)$, then we may write

$$K_Y + \Gamma_t = \pi^*(K_X + \Delta_t) + E_t$$

as in (*) of §1.1. We may assume that there is an effective and exceptional $\mathbb{Q}$ -divisor F such that $f^*A' - F$ is ample. Let $C \sim_{\mathbb{Q}} f^*A' - F$ be a general ample $\mathbb{Q}$ -divisor and define

$$\Gamma'_t = \Gamma_t - f^*A' + F + C \sim_{\mathbb{Q}} \Gamma_t, \quad \Gamma''_t = \Gamma'_t - (\Gamma'_t \wedge N_{\sigma}(K_Y + \Gamma'_t)).$$

The following properties now hold:

- (1) (Y, Γ''_0) is kawamata log terminal;
- (2) $\Gamma''_t \geq C$ for any $t \in [0, u]$;
- (3) (Y, Γ''_u) is not log canonical;
- (4) Y is smooth and Γ''_t has simple normal crossings support for any $t \in [0, u]$;
and
- (5) $\Gamma''_t \wedge N_{\sigma}(K_Y + \Gamma''_t) = 0$ for any $t \in [0, u]$.

Let $s = \sup\{t \in [0, u] \mid K_Y + \Gamma''_t \text{ is LC}\}$. Then $\Gamma''_s = (f^{-1})_*S + C + B'$ is purely log terminal. We may therefore assume that (X, Δ) satisfies properties (1) and (3). Property (2) may now be achieved as follows. Let $f : Y \rightarrow X$ be a log resolution of (X, Δ) such that there is an effective and exceptional $\mathbb{Q}$ -divisor F such that $f^*A - F$ is ample. Let $C \sim_{\mathbb{Q}} f^*A - F$ be a general ample $\mathbb{Q}$ -divisor. We may write

$$K_Y + \Gamma = f^*(K_X + \Delta) + E$$

as in (*) of §1.1. Let $T = f_*^{-1}S$. We may assume that $K_T + \Gamma_T = (K_Y + \Gamma)|_T$ is terminal. Let $\Gamma' = \Gamma - f^*A + F + C$. We may then also assume that $K_T + \Gamma'_T + C|_T = (K_Y + \Gamma' + C)|_T$ is terminal. After replacing Γ' by $\Gamma' - (\Gamma' \wedge N_{\sigma}(K_Y + \Gamma'))$, properties (1), (2) and (3) are satisfied.

Step 5. The theorem now follows from (7.16) below. $\square$

Theorem 7.16. *Let $(X, \Delta = S + A + B)$ be a projective purely log terminal log smooth pair where A is a general ample $\mathbb{Q}$ -divisor, $(S, \Omega + A|_S)$ is terminal where $\Omega = (A + B)|_S$, $B \geq 0$ and $S = \lfloor \Delta \rfloor$ is irreducible. If $K_X + \Delta$ is pseudo-effective and S is not contained in $N_{\sigma}(K_X + \Delta)$ then $K_X + \Delta \sim_{\mathbb{R}} D \geq 0$.*

Proof. Since $K_X + \Delta$ is pseudo-effective and S is not contained in $N_{\sigma}(K_X + \Delta)$, by (7.13), S is not contained in $\mathbf{B}_{-}(K_X + \Delta)$ and so $N_{\sigma}(\|K_X + \Delta\|_S)$ is defined. (Note that as observed above, the support of $N_{\sigma}(\|K_X + \Delta\|_S)$ consist of the union of countably many prime divisors in X .) Let

$$\Phi = \sum_{P \in \text{Supp}(\Omega)} \sigma_P(\|K_X + \Delta\|_S) \cdot P \leq N_{\sigma}(\|K_X + \Delta\|_S)$$

and $\Theta = \Omega - \Omega \wedge \Phi$. By (7.13), $K_S + \Theta$ is pseudo-effective.

Let $W \subset \text{WDiv}_{\mathbb{R}}(X)$ be the smallest rational affine subspace containing Δ , $V \subset \text{WDiv}_{\mathbb{R}}(S)$ be the smallest rational affine subspace containing Θ and $U \subset \text{WDiv}_{\mathbb{R}}(S)$ be the smallest rational affine subspace containing Φ .

There exists a number $\epsilon > 0$ such that $\epsilon(K_X + \Delta) + \frac{1}{2}A$ is ample and if $\Delta' \in W$, $\|\Delta - \Delta'\| < \epsilon$, then $\Delta' - \Delta + \frac{1}{2}A$ is ample.

By compactness and (2.11), there is a real number $\eta > 0$ and an integer $r > 0$ such that if $\Theta' \in V$ and $\|\Theta - \Theta'\| < \eta$, then $K_S + \Theta'$ is pseudo-effective and if $k > 0$ is an integer such that $k(K_S + \Theta')/r$ is Cartier, then $h^0(\mathcal{O}_S(k(K_S + \Theta')) > 0$.

Let W' be the smallest rational affine subspace of $W \times U$ containing (Δ, Φ) . In particular W' dominates W and U . By Diophantine Approximation, there exist $\mathbb{Q}$ -divisors $(\Delta_i, \Phi_i) \in W'$ and integers $k_i \gg 0$ such that:

- (1) we may write $\Delta = \sum r_i \Delta_i$ where $r_i > 0$, $\sum r_i = 1$,
- (2) $(X, \Delta_i = S + A + B_i)$ is purely log terminal and $(S, \Omega_i + A|_S)$ is terminal where $\Omega_i = (A + B_i)|_S$,
- (3) $k_i \Delta_i / r$ are integral and $\|\Delta - \Delta_i\| < \epsilon / k_i$,
- (4) $k_i \Phi_i / r$ are integral and $\|\Phi - \Phi_i\| < \phi \epsilon / k_i$ where ϕ is the minimum non-zero coefficient of Φ .

Claim 7.17. *We may assume that $\Theta_i = \Omega_i - (\Omega_i \wedge \Phi_i)$ belongs to V .*

Proof of Claim. Let P be a prime divisor on S contained in $\text{Supp}(\Omega)$. Notice that if $\text{mult}_P((\Delta - S)|_S) = \text{mult}_P(\Phi)$, then $\text{mult}_P((\Delta' - S)|_S) = \text{mult}_P(\Phi')$ for any $(\Delta', \Phi') \in W'$. Let $(\Delta', \Phi') \in W'$. If $\|\Delta - \Delta'\| \ll 1$ and $\|\Phi - \Phi'\| \ll 1$, then $\text{mult}_P((\Delta - S)|_S) > \text{mult}_P(\Phi)$ if and only if $\text{mult}_P((\Delta' - S)|_S) > \text{mult}_P(\Phi')$. It follows that the assignment $\theta : W' \rightarrow \text{WDiv}_{\mathbb{R}}(S)$ given by

$$\theta(\Delta', \Phi') = (\Delta' - S)|_S - (\Delta' - S)|_S \wedge \Phi'$$

is rational linear and $\theta(\Delta, \Phi) = \Theta \in V$. Since V is a rational affine space, $\theta^{-1}(V) \subset W'$ is a rational affine subspace of W' containing (Δ, Φ) and so $\theta^{-1}(V) = W'$ and hence $V = \theta(W')$. $\square$

Claim 7.18. *Let $P \in \text{Supp}(\Omega)$, then for any $l > 0$ sufficiently divisible,*

$$\text{mult}_P(\text{Fix}|l(K_X + \Delta_i + \frac{1}{k_i}A)|_S) \leq l \text{mult}_P(\Phi_i).$$

Proof of Claim. Since $\|\Phi - \Phi_i\| < \phi \epsilon / k_i$, it suffices to show that

$$\text{mult}_P(\text{Fix}|l(K_X + \Delta_i + \frac{1}{k_i}A)|_S) \leq l(1 - \frac{\epsilon}{k_i}) \text{mult}_P(\Phi).$$

Since $\|\Delta_i - \Delta\| \ll 1$, $\epsilon(K_X + \Delta_i) + \frac{1}{2}A$ is ample. Let $\eta > \epsilon / k_i$ be a rational number such that $\eta(K_X + \Delta_i) + \frac{1}{2k_i}A$ is ample. Notice that

$$l(K_X + \Delta_i + \frac{1}{k_i}A) = l(1 - \eta)(K_X + \Delta_i + \frac{1}{2k_i}A) + l(\eta(K_X + \Delta_i) + \frac{1 + \eta}{2k_i}A).$$

Since $\Delta_i - \Delta + \frac{1}{2k_i}A$ is ample, we may assume that $\text{Fix}|l(1 - \eta)(K_X + \Delta_i + \frac{1}{2k_i}A)|_S \leq l(1 - \eta)\Phi$ cf. (7.14). Since $\eta(K_X + \Delta_i) + \frac{1 + \eta}{2k_i}A$ is ample, the assertion follows. $\square$

Claim 7.19. *There exists an effective divisor H on X not containing S such that for all sufficiently divisible integers $m > 0$, we have*

$$|m(K_S + \Theta_i)| + mF_i + (\frac{m}{k_i}A + H)|_S \subset |m(K_X + \Delta_i) + \frac{m}{k_i}A + H|_S,$$

where $F_i = \Omega_i - \Theta_i$.

Proof of Claim. Since $S \not\subset N_\sigma(K_X + \Delta)$ and $\Delta_i - \Delta + \frac{1}{k_i}A$ is ample, it follows that $S \not\subset \text{Bs}(m(K_X + \Delta_i + \frac{1}{k_i}A))$ for m sufficiently divisible cf. (7.5). Let $f : Y \rightarrow X$ be a log resolution of $(X, \Delta_i + \frac{1}{k_i}A)$ and of $|m(K_X + \Delta_i + \frac{1}{k_i}A)|$. We write

$$K_Y + \Gamma = f^*(K_X + \Delta_i + \frac{1}{k_i}A) + E$$

as in (*) of §1.1. Define

$$\Xi = \Gamma - \Gamma \wedge \frac{\text{Fix}(m(K_Y + \Gamma))}{m}.$$

We have that $m(K_Y + \Xi)$ is Cartier, $\text{Fix}(m(K_Y + \Xi)) \wedge \Xi = 0$ and $\text{Mob}(m(K_Y + \Xi))$ is free. Since $\text{Fix}(m(K_Y + \Xi)) + \Xi$ has simple normal crossings support, it follows that $\mathbf{B}(K_Y + \Xi)$ contains no log canonical center of $(Y, \Gamma \Xi^\vee)$. Let $T = (f^{-1})_* S$. Let $\Gamma_T = (\Gamma - T)|_T$ and $\Xi_T = (\Xi - T)|_T$. Let $m > 0$ be divisible by l . By (4.16)

$$\sigma \in H^0(T, \mathcal{O}_T(m(K_T + \Xi_T))) = H^0(T, \mathcal{J}_{\|m(K_T + \Xi_T)\|}(m(K_T + \Xi_T))).$$

By (4.11), there is an ample divisor H on Y such that if $\tau \in H^0(T, \mathcal{O}_T(H))$, then $\sigma \cdot \tau$ is in the image of the homomorphism

$$H^0(Y, \mathcal{O}_Y(m(K_Y + \Xi) + H)) \rightarrow H^0(T, \mathcal{O}_T(m(K_Y + \Xi) + H)).$$

Therefore

$$(\dagger) \quad |m(K_T + \Xi_T)| + m(\Gamma_T - \Xi_T) + |H|_T \subset |m(K_Y + \Gamma) + H|_T.$$

We claim that

$$(b) \quad \Omega_i + \frac{1}{k_i} A|_S \geq (f|_T)_* \Xi_T \geq \Omega_i - (\Phi_i \wedge \Omega_i) + \frac{1}{k_i} A|_S$$

and so, as $(S, \Omega_i + \frac{1}{k_i} A|_S)$ is canonical, we have

$$|m(K_S + \Omega_i - (\Phi_i \wedge \Omega_i))| + m((f|_T)_* \Xi_T - (\Omega_i - (\Phi_i \wedge \Omega_i))) \subset |m(K_S + (f|_T)_* \Xi_T)| = (f|_T)_* |m(K_T + \Xi_T)|$$

cf. (4.24). Pushing forward the inclusion $(\dagger)$, one sees that

$$|m(K_S + \Omega_i - (\Phi_i \wedge \Omega_i))| + m(\Phi_i \wedge \Omega_i) + \left(\frac{m}{k_i} A + f_* H\right)|_S \subset |m(K_X + \Delta_i) + \frac{m}{k_i} A + f_* H|_S.$$

The Claim now follows.

We will now prove the inequality (b) claimed above. We have $\Xi_T \leq \Gamma_T$ and $(f|_T)_* \Gamma_T = \Omega_i + \frac{1}{k_i} A|_S$ and so the first inequality follows.

In order to prove the second inequality, let P be any prime divisor on S and let P' be its strict transform on T . Assume that P is contained in the support of Ω_i , or equivalently, that P' is contained in the support of $(\Gamma - T)|_T$. Then there is a component G of the support of Γ such that

$$\text{mult}_{P'}(\text{Fix}|m(K_Y + \Gamma)|_T) = \text{mult}_G(\text{Fix}|m(K_Y + \Gamma)|)$$

and $\text{mult}_G(\Gamma) = \text{mult}_{P'}(\Gamma_T)$. Therefore

$$\text{mult}_{P'}(\Xi_T) = \text{mult}_G(\Xi) = \text{mult}_{P'}(\Gamma_T) - \min\{\text{mult}_{P'}(\Gamma_T), \frac{1}{m} \text{mult}_{P'} \text{Fix}|m(K_Y + \Gamma)|_T\}.$$

Notice that $\text{mult}_P(\Omega_i + \frac{1}{k_i} A|_S) = \text{mult}_{P'}(\Gamma_T)$ and since $E|_T$ is exceptional, we have that

$$\text{mult}_{P'}(\text{Fix}|m(K_Y + \Gamma)|_T) = \text{mult}_P(\text{Fix}|m(K_X + \Delta_i + \frac{1}{k_i} A)|_S).$$

Therefore $(f|_T)_* \Xi_T = \Omega_i + \frac{1}{k_i} A|_S - \Omega_i \wedge \frac{1}{m} \text{Fix}|m(K_X + \Delta_i + \frac{1}{k_i} A)|_S$. The inequality now follows from (7.18). $\square$

Claim 7.20. *We have*

$$|k_i(K_S + \Theta_i)| + k_i F_i \subset |k_i(K_X + \Delta_i)|_S.$$

Proof of claim. For any $\Sigma \in |k_i(K_S + \Theta_i)|$ and any $m > 0$ sufficiently divisible, we may choose a divisor $G \in |m(K_X + \Delta_i) + \frac{m}{k_i}A + H|$ such that $G|_S = \frac{m}{k_i}\Sigma + mF_i + (\frac{m}{k_i}A + H)|_S$. If we define $\Lambda = \frac{k_i-1}{m}G + B_i$, then

$$k_i(K_X + \Delta) \sim_{\mathbb{Q}} K_X + S + \Lambda + (\frac{1}{k_i}A - \frac{k_i-1}{m}H)$$

where $\frac{1}{k_i}A - \frac{k_i-1}{m}H$ is ample as m is sufficiently big. By (4.6), we have a surjective homomorphism

$$H^0(X, \mathcal{J}_{S,\Lambda}(k_i(K_X + \Delta))) \rightarrow H^0(S, \mathcal{J}_{\Lambda|_S}(k_i(K_X + \Delta))).$$

Since (S, Ω_i) is canonical, $(S, \Omega_i + \frac{k_i-1}{m}H|_S)$ is kawamata log terminal as m is sufficiently big. Therefore $\mathcal{J}_{\Omega_i + \frac{k_i-1}{m}H|_S} = \mathcal{O}_S$. Since

$$\Lambda|_S - (\Sigma + k_i\Phi) = (\frac{k_i-1}{m}G + B)|_S - (\Sigma + k_iF_i) \leq \Omega_i + \frac{k-1}{m}H|_S,$$

then by (2) of (4.4), we have $\mathcal{I}_{\Sigma + k_iF_i} \subset \mathcal{J}_{\Lambda|_S}$ and so

$$\Sigma + k_iF_i \in |k_i(K_X + \Delta_i)|_S$$

and the claim follows. $\square$

Note that as $\Theta_i \in V$ and $\|\Theta - \Theta_i\| < \eta$, $h^0(\mathcal{O}_S(k_i(K_S + \Theta_i))) > 0$ and so by (7.20), there exists an effective divisor D_i such that $k_i(K_X + \Delta_i) \sim_{\mathbb{Q}} D_i$. But then

$$K_X + \Delta \sim_{\mathbb{R}} \sum r_i D_i \geq 0.$$

$\square$

8. FINITENESS OF LOG TERMINAL MODELS

Theorem 8.1. *Theorems D, E in dimension $\leq n-1$ and Theorems A, B, C in dimension n imply Theorem D (1) in dimension n .*

Proof. We proceed by induction on the dimension of $\mathcal{C}$. The case when $\dim \mathcal{C} = 0$ follows immediately from Theorems B and C in dimension n .

Notice that $\text{PSEF}(X/U)$ (the cone of pseudo-effective divisors over U) is closed. By compactness of $\mathcal{C} \cap \text{PSEF}(X/U)$, it suffices to work locally around an $\mathbb{R}$ -divisor $K_X + \Delta_0 \in \mathcal{C} \cap \text{PSEF}(X/U)$. We are therefore free to replace $\mathcal{C}$ by an appropriate neighborhood of $K_X + \Delta_0$ inside $\mathcal{C}$.

Step 1. *We may assume that $K_X + \Delta_0$ is nef over U .*

By Theorems B and C in dimension n , there exists a log terminal model $\phi : X \dashrightarrow Y$ for $K_X + \Delta_0$ over U . Note that if $K_X + \Delta$ is pseudo-effective over U , then $K_Y + \phi_*\Delta$ is pseudo-effective over U . After shrinking $\mathcal{C}$, we have for all $K_X + \Delta \in \mathcal{C}$ that:

- (1) $K_Y + \phi_*\Delta$ is kawamata log terminal cf. (8.4), and
- (2) $a_F(X, \Delta) < a_F(Y, \phi_*\Delta)$ for any ϕ -exceptional divisor $F \subset X$.

It follows that if $\psi : Y \rightarrow Z$ is a log terminal model for $(Y, \phi_*\Delta)$ over U , then $\psi \circ \phi : X \rightarrow Z$ is a log terminal model for (X, Δ) over U cf. (8.5). We may therefore assume that $K_X + \Delta_0$ is nef over U .

Step 2. *We may assume that $K_X + \Delta_0 \sim_{\mathbb{R}, U} 0$.*

Since $K_X + \Delta_0$ is nef over U and Δ_0 is big over U , by the Base Point Free Theorem (cf. (1.14)), $K_X + \Delta_0$ is semiample over U and so there is a morphism $\psi : X \rightarrow Z$ over U and an $\mathbb{R}$ -divisor H on Z ample over U such that $K_X + \Delta_0 \sim_{\mathbb{R}, U} \psi^*H$.

After further shrinking $\mathcal{C}$, we may assume that any log terminal model for (X, Δ) over Z is a log terminal model for (X, Δ) over U cf. (8.6). We may hence replace U by Z so that we may assume that $K_X + \Delta_0 \sim_{\mathbb{R}, U} 0$.

Step 3. Pick any $K_X + \Theta \in \mathcal{C}$ and let $K_X + \Delta$ be the point on the boundary of $\mathcal{C}$ such that

$$\Theta - \Delta_0 = \lambda(\Delta - \Delta_0) \quad 0 < \lambda < 1.$$

Then we have that

$$K_X + \Theta = \lambda(K_X + \Delta) + (1 - \lambda)(K_X + \Delta_0) \sim_{\mathbb{R}, U} \lambda(K_X + \Delta).$$

It follows that $K_X + \Delta$ is pseudo-effective over U if and only if $K_X + \Theta$ is pseudo-effective over U and $\phi : X \dashrightarrow Y$ is a log terminal model for (X, Δ) over U if and only if it is a log terminal model for (X, Θ) over U . Since the boundary of $\mathcal{C}$ is a rational polytope of strictly smaller dimension, the induction is now complete. $\square$

Theorem 8.2. *Theorems D, E in dimension $\leq n - 1$ and Theorems A, B, C in dimension n imply Theorem D in dimension n .*

Proof. Notice that if $\phi : X \dashrightarrow Y$ is any log terminal model of (X, Δ) over U and $H \in \text{WDiv}_{\mathbb{R}}(Y)$ is ample over U , then ϕ is a log terminal model over U for any kawamata log terminal pair (X, Δ') such that $\Delta' \equiv_U \Delta + \epsilon \phi_*^{-1} H$ and $0 < \epsilon \ll 1$. Since $K_Y + \phi_* \Delta'$ is ample over U , ϕ is in fact the unique log terminal model over U for (X, Δ') .

It suffices therefore to show that there exists a rational polytope $\mathcal{C} \subset \mathcal{C}'$ contained in a finite dimensional subspace of $\text{WDiv}_{\mathbb{R}}(X)$ such that if $K_X + \Delta \in \mathcal{C}$ and H' is an $\mathbb{R}$ -Cartier divisor on X , then there exists a kawamata log terminal pair $K_X + \Delta' \in \mathcal{C}'$ and a constant $\epsilon > 0$ such that $\Delta' \equiv_U \Delta + \epsilon H'$. This can be achieved as follows: First, we translate $\mathcal{C}$ by $A' + B' - A$ where A' is a general ample $\mathbb{Q}$ -divisor over U , $B' \geq 0$ is a $\mathbb{Q}$ -divisor such that $A \sim_{\mathbb{R}, U} A' + B'$, the components of $\text{Supp}(B')$ generate $\text{WDiv}_{\mathbb{R}}(X)/\equiv_U$, and for any $K_X + \Delta \in \mathcal{C}$, $\Delta - A + A' + B'$ is effective and $(X, \Delta - A + A' + B')$ is kawamata log terminal. We then let

$$\mathcal{C}' = \{K_X + \Delta' = K_X + \Delta + C \mid K_X + \Delta \in \mathcal{C}, \text{Supp}(C) \subset \text{Supp}(B') \text{ and } \|C\| < \epsilon\}$$

where $0 < \epsilon \ll 1$. $\square$

Theorem 8.3. *Theorem E in dimension $\leq n - 1$ and Theorems A, B, C, D in dimension n imply Theorem E in dimension n .*

Proof. (1) is immediate from (2.7) and (1.14). We now prove (2). Notice that if $K_X + \Delta'$ is not pseudo-effective, then $\mathbf{B}(K_X + \Delta') = X$. Therefore, we may assume that $K_X + \Delta'$ is pseudo-effective. Let $\phi : X \dashrightarrow Y$ be a log terminal model for $K_X + \Delta$ over Z . We may pick $0 < \delta$ such that if $\|\Delta - \Delta'\| \leq \delta$, then $K_Y + \phi_* \Delta'$ is kawamata log terminal cf. (8.4) and $a_F(X, \Delta') < a_F(Y, \phi_* \Delta')$ for any ϕ -exceptional divisor $F \subset X$. By (8.5), if $\psi : Y \rightarrow Z$ is a log terminal model for $K_Y + \phi_* \Delta'$ over Z , then $\psi \circ \phi$ is a log terminal model for $K_X + \Delta'$ over Z . By (2.21), all ϕ -exceptional divisors are contained in $\mathbf{B}(K_X + \Delta')$.

We now prove (3). Let $\phi : X \dashrightarrow Y$ be a log terminal model for $K_X + \Delta$ over Z and let $\mathcal{C}$ be the set of all $\Delta' \in W$ such that $\|\Delta - \Delta'\| \leq \eta$ and ϕ is a log terminal model for $K_X + \Delta'$ over Z . Since Δ is big, we may assume that there is $\mathbb{Q}$ -divisor A on Y ample over Z such that if $\|\Delta - \Delta'\| \leq \eta$, then $\phi_* \Delta' \geq A$. If η

is sufficiently small, then $\phi_*(\Delta' - \Delta) + A/2$ is ample over Z for any $\Delta' \in W$ such that $\|\Delta - \Delta'\| \leq \eta$ and moreover

$$\mathcal{C} = \{\Delta' \in W \mid K_Y + \phi_*\Delta' \text{ is nef over } Z, \text{ and } \|\Delta - \Delta'\| \leq \eta\}.$$

By the Cone Theorem cf. (1.7), there are finitely many negative extremal rays $R_1, \dots, R_k$ for $K_Y + \phi_*\Delta - A/2$. Since

$$K_Y + \phi_*\Delta' = (K_Y + \phi_*\Delta - A/2) + \phi_*(\Delta' - \Delta) + A/2,$$

if $K_Y + \phi_*\Delta'$ is not nef over Z , then there is an integer $1 \leq i \leq k$ such that R_i is a negative extremal ray for $K_Y + \phi_*\Delta'$ over Z . Therefore $\mathcal{C}$ is cut out by a subset of the finite collection of extremal rays $R_i = \mathbb{R}^+[C_i]$ and hence $\mathcal{C}$ is a rational polytope. Therefore the affine subspace of $\text{WDiv}_{\mathbb{R}}(X)$ generated by $\mathcal{C}$ is W . We may then assume that $\mathcal{C} = \{\Delta' \in W : \|\Delta - \Delta'\| \leq \eta\}$.

Since Y is $\mathbb{Q}$ -factorial, there is an integer $l > 0$ such that if $G \in \text{WDiv}(Y)$, then lG is Cartier. By [Kollár93], there exists an integer $r > 0$ such that if $k(K_Y + \phi_*\Delta')/r$ is integral, then $k(K_Y + \phi_*\Delta')$ is generated by global sections. Let $p : W \rightarrow X$ and $q : W \rightarrow Y$ be a common resolution, then by definition of log terminal model, $p^*(K_X + \Delta') = q^*(K_Y + \phi_*\Delta') + E$ where $E \geq 0$ is q -exceptional and the support of $p_*(E)$ is the union of all ϕ -exceptional divisors on X . It follows that the support of $\text{Fix}(k(K_X + \Delta'))$ is given by the divisors contained in $\mathbf{B}(K_X + \Delta')$ i.e. by the divisors contained the support of $p_*(E)$. Since ϕ is a log terminal model for $K_X + \Delta$ over Z , these divisors are also contained in $\mathbf{B}(K_X + \Delta)$. $\square$

Exercise 8.4. Let (X, Δ_0) be a $\mathbb{Q}$ -factorial kawamata log terminal pair and $\Delta_0 \in V \subset \text{WDiv}_{\mathbb{R}}(X)$ a finite dimensional vector space. Show that there exists an $\epsilon > 0$ such that $0 \leq \Delta \in V$ and $\|\Delta\| \leq \epsilon$, then (X, Δ) is kawamata log terminal.

Exercise 8.5. Let (X, Δ) be a $\mathbb{Q}$ -factorial kawamata log terminal pair $\pi : X \rightarrow U$ a projective morphism, $\phi : X \dashrightarrow Y$ be a rational map over U that extracts no divisors such that Y is normal, $\mathbb{Q}$ -factorial and projective over U . If $a_F(X, \Delta) < a_F(Y, \phi_*\Delta)$ for any ϕ -exceptional divisor $F \subset X$ and if $\psi : Y \dashrightarrow Z$ is a log terminal model of $(Y, \phi_*\Delta)$ over U , then $\psi \circ \phi$ is a log terminal model of (X, Δ) over U .

Exercise 8.6. Let $\psi : X \rightarrow Z$ and $\eta : Z \rightarrow U$ be projective morphisms of normal varieties. If H is an ample $\mathbb{R}$ -divisor on Z and (X, Δ) is a $\mathbb{Q}$ -factorial kawamata log terminal pair such that Δ is big over U and $K_X + \Delta \sim_{\mathbb{R}, U} \psi^*H$, show that there exists $0 < \epsilon \ll 1$ such that if $\|\Delta - \Delta_0\| \leq \epsilon$, then any log terminal model for (X, Δ) over Z is a log terminal model for (X, Δ) over U .

9. SOLUTIONS TO SOME OF THE EXERCISES

Hint to (1.8). The inclusion $\mathbf{B}(D) \subset \text{Bs}(mD)$ is clear. If $x \notin \mathbf{B}(D)$, then there is an $\mathbb{R}$ -divisor $D' \geq 0$ such that $D \sim_{\mathbb{R}} D'$ and $x \notin \text{Supp}(D')$. We have $D' - D = \sum r_i(f_i)$ where $r_i \in \mathbb{R}$ and f_i are rational functions on X . We may assume that D and D' have no common components and that $\text{Supp}(f_i) \subset \text{Supp}(D + D')$ (eg. by assuming that r_i are linearly independent over $\mathbb{Q}$). Then if q_i are rational numbers with $|q_i - r_i| \ll 1$, we have that $D'' = D + \sum q_i(f_i)$ is a $\mathbb{Q}$ -divisor $\mathbb{Q}$ -linearly equivalent to D with the same support as D' . $\square$

Hint to (1.10). Blow up smooth divisors contained in F and its strict transforms. $\square$

Solution to (1.11). Let $A = L - (K_Y + \Gamma)$. There exists an effective divisor $\Gamma' \sim_{\mathbb{R}, X} \Gamma + A/2$ such that (Y, Γ') is kawamata log terminal and $L - (K_Y + \Gamma') \sim_{\mathbb{R}, X} A/2$ is f -ample. $\square$

Solution to (1.12). [KM98, 5.48, 5.50]. $\square$

Solution to (1.13). [KM98, 2.43]. $\square$

Solution to (1.14). We may write $A \sim_{\mathbb{R}} A' + C$ where A' is a general ample $\mathbb{Q}$ -divisor and C is an effective $\mathbb{R}$ -divisor whose support contains no log canonical center of (X, Δ) . For $0 < \epsilon \ll 1$, we then have that $(X, \Delta' = (1 - \epsilon)\Delta + \epsilon C)$ is kawamata log terminal cf. (1.13), $D = K_X + \Delta$ is nef and $D - (K_X + \Delta')$ is ample. The claim now follows from (1.6). $\square$

Solution to (2.17). If $K_Z + f_*\Delta$ is $\mathbb{Q}$ -Cartier, then as f is small, $K_X + \Delta = f^*(K_Z + f_*\Delta)$. This contradicts the fact that $-(K_X + \Delta)$ is ample over Z . $\square$

Solution to (2.18). See [KM98, 3.37, 3.38, 3.42]. $\square$

Solution to (2.19). See [KM98, 3.36, 3.43]. $\square$

Solution to (2.20). By (2.18) and (2.19), the X_i are $\mathbb{Q}$ -factorial. It suffices to prove that $R(K_{X_i} + \Delta_i) \cong R(K_{X_{i+1}} + \Delta_{i+1})$. Let $p : W \rightarrow X_i$ and $q : W \rightarrow X_{i+1}$ be a common resolution that resolves the indeterminacies of ϕ_i . By (2.18) and (2.19), we have that

$$p^*(K_{X_i} + \Delta_i) = q^*(K_{X_{i+1}} + \Delta_{i+1}) + F$$

where $F \geq 0$ is a q -exceptional $\mathbb{Q}$ -divisor. $\square$

Solution to (2.21). Let $\phi : X \dashrightarrow Y$ be a log terminal model for (X, Δ) over Z . Then $(Y, \phi_*\Delta)$ is kawamata log terminal and Δ is big. By the Base Point Free Theorem, it follows that $K_Y + \phi_*\Delta$ is semiample over Z . Let $p : W \rightarrow X$ and $q : W \rightarrow Y$ be a common resolution that resolves the indeterminacies of ϕ . Then

$$p^*(K_X + \Delta) = q^*(K_Y + \phi_*\Delta) + F$$

where $F \geq 0$ is a q -exceptional $\mathbb{Q}$ -divisor whose support contains the divisors contracted by ϕ . The assertion now follows easily. $\square$

Solution to (2.22). Let $\phi : X \dashrightarrow Y$ be a log terminal model for $K_X + \Delta$ over Z , then $(Y, \phi_*\Delta)$ is kawamata log terminal and $\phi_*\Delta$ is big over Z therefore the same is true for $(Y, \phi_*\Delta')$. Therefore, there is a log terminal model for $K_Y + \phi_*\Delta'$ over Z , say $\psi : Y \dashrightarrow W$. It is easy to see that $\psi \circ \phi$ is a log terminal model for $K_X + \Delta'$ over Z . The claim now follows from (2.21). $\square$

Solution to (3.10). Let $\Delta = S + \sum \delta_i \Delta_i$ and set $\Delta' = S + \sum \delta'_i \Delta_i$ where $0 < \delta'_i < \delta_i$ are rational numbers such that $|\delta'_i - \delta_i| \ll 1$. It is easy to check that f is a pl-flipping contraction for (X, Δ') . $\square$

Solution to (3.11). Since Z is affine, the sheaf $f_*\mathcal{O}_X(S)$ is generated by global sections. Let $S' \sim S$ be a divisor corresponding to a general section of $\Gamma(Z, f_*\mathcal{O}_X(S)) \cong \Gamma(X, \mathcal{O}_X(S))$. Since f is an isomorphism in codimension 1 and $\mathcal{O}_X(S)$ is invertible in codimension 1, S and S' have no common components. $\square$

Solution to (4.13). Let $f: Y \rightarrow X$ be a log resolution of (X, Δ) and V , then $M = f^*D - \text{Fix}(f^*V)$ is a reduced irreducible smooth divisor intersecting $F = \text{Fix}(f^*V)$ transversely. Since $c < 1$, we have

$$\mathcal{J}_{\Delta, c \cdot V} = f_* \mathcal{O}_Y(E - \lfloor c \cdot F \rfloor) = f_* \mathcal{O}_Y(E - \lfloor c \cdot (M + F) \rfloor) = \mathcal{J}_{\Delta, c \cdot D}.$$

□

Solution to (4.14). Let $f: Y \rightarrow X$ be a log resolution of $(X, \Delta + D)$ that is an isomorphism at the general point of each log canonical center of (X, Δ) . If $\mathcal{J}_{\Delta, D} = \mathcal{O}_X$, then $E - \lfloor f^*D \rfloor \geq 0$ and in particular $\lfloor D \rfloor = 0$. For any exceptional prime divisor F in Y , we have $a_F(X, \Delta + D) = \text{mult}_F(E - f^*D) > -1$ so that $(X, \Delta + D)$ is divisorially log terminal. The reverse implication is similar. □

Solution to (4.16). Let $p > 0$ be an integer such that $\mathcal{J}_{\lfloor D \rfloor} = \mathcal{J}_{\frac{1}{p} \cdot \lfloor pD \rfloor}$. Let $f: Y \rightarrow X$ be a log resolution of $\lfloor pD \rfloor$. Then $\mathcal{J}_{\lfloor D \rfloor} = f_* \mathcal{O}_Y(K_{Y/X} - \lfloor \frac{1}{p} F_p \rfloor)$ where $F_p = \text{Fix}(\lfloor pf^*D \rfloor)$. If $F_1 = \text{Fix}(\lfloor f^*D \rfloor)$, then $pF_1 \geq F_p$ and so

$$\begin{aligned} H^0(\mathcal{O}_X(D)) &= H^0(\mathcal{O}_Y(f^*D - F_1)) \subset H^0(\mathcal{O}_Y(f^*D + K_{Y/X} - \lfloor \frac{1}{p} F_p \rfloor)) \\ &= H^0(\mathcal{J}_{\lfloor D \rfloor}(D)) \subset H^0(\mathcal{O}_X(D)). \end{aligned}$$

□

Solution to (4.15). Let $f: Y \rightarrow X$ be a log resolution of $(X, \Delta + D)$. Since $E \geq 0$, we have that $E - \lfloor c \cdot f^*D \rfloor$ is effective and f -exceptional for $0 < c \ll 1$. Therefore $\mathcal{J}_{\Delta, c \cdot D} = \mathcal{O}_X$.

If $\text{mult}_x(D) > 1$, let $\mu: X' \rightarrow X$ be the blow up of $x \in X$ with exceptional divisor F . Then $\text{mult}_F(K_{X'} + (\mu^{-1})^*D - \mu^*(K_X + D)) \geq 1$. It follows that for any log resolution $f: Y \rightarrow X$ dominating μ , we have that $E - \lfloor f^*D \rfloor$ is not effective and in fact contained in $\mathcal{O}_Y(-F')$ where F' is the strict transform of F . Therefore $\mathcal{J}_{\Delta, D} \subset f_* \mathcal{O}_Y(-F') = m_x$. □

Solution to (4.17). For any $m > 0$, there exists a number $k > 0$ such that $\text{mult}_Z(\lfloor mkD \rfloor)/k < 1$. By [Lazarsfeld04, 9.5.13], we have that $\mathcal{J}_{\frac{1}{k} \cdot \lfloor mkD \rfloor} = \mathcal{O}_X$ on a neighborhood of a general point of Z . Let H be an ample $\mathbb{Q}$ -divisor on X . For any $m > 0$ sufficiently big and divisible, $\mathcal{O}_X(m(D + H)) \otimes \mathcal{J}_{\frac{1}{k} \cdot \lfloor mkD \rfloor}$ is globally generated cf. (3) of (4.10). □

Solution to (4.19). Let $f: Y \rightarrow X$ be a log resolution of $(X, \Delta + D)$. By the projection formula, we have

$$\begin{aligned} \mathcal{J}_{\Delta, D} &= f_* \mathcal{O}_Y(E - \lfloor f^*(\lfloor D \rfloor + \{D\}) \rfloor) = \\ &= f_* \mathcal{O}_Y(E - \lfloor f^*\{D\} \rfloor) \otimes \mathcal{O}_X(-\lfloor D \rfloor) = \mathcal{J}_{\Delta, \{D\}} \otimes \mathcal{O}_X(-\lfloor D \rfloor). \end{aligned}$$

□

Solution to (4.20). Let H be an ample divisor on X and $H' = f_*H$. Then $F = f^*H' - H$ is effective and $-F$ is ample over Y . □

Solution to (4.21). Since the spectral sequence $E_{p,q}^1 = R^p \pi_* R^q f_* F$ degenerates at the E^1 -term since $E_{p,q}^1 = 0$ for $q > 0$. □

Solution to (4.22). See [Lazarsfeld04, 1.8.5]. □

Solution to (4.24). Since $(X, f_*\Xi)$ is canonical, $K_Y + \Xi = f^*(K_X + f_*\Xi) + E$ where E is effective and exceptional. $\square$

Solution to (8.4). Let G be a divisor whose support contains the support of any effective divisor in V and let $f : Y \rightarrow X$ be a log resolution of (X, G) . It follows that f is a log resolution of (X, Δ) for any $0 \leq \Delta \in V$. We let $K_Y + \Gamma = f^*(K_X + \Delta)$. (X, Δ) is kawamata log terminal if $\text{mult}_F(\Gamma) < 1$ where F is any divisor in the support of G or any f -exceptional divisor on Y . Since (X, Δ_0) is kawamata log terminal and $\text{mult}_F(\Gamma)$ are continuous functions of Δ , the assertion is clear. $\square$

Solution to (8.5). By the Negativity Lemma, it suffices to check that $a_F(X, \Delta) < a_F(X, \psi_*\phi_*\Delta)$ for all $(\psi \circ \phi)$ -exceptional divisors $F \subset X$. Let $p : W \rightarrow X$, $q : W \rightarrow Y$ and $r : W \rightarrow Z$ be a common resolution, then we have that $q^*(K_Y + \phi_*\Delta) = r^*(K_Z + \psi_*\phi_*\Delta) + E$ where $E \geq 0$ is r -exceptional and that $K_X + \Delta = p_*q^*(K_Y + \phi_*\Delta) + G$ where $G \geq 0$ is ϕ -exceptional. Therefore $K_X + \Delta = p_*r^*(K_Z + \psi_*\phi_*\Delta) + p_*E + G$ where $p_*E + G \geq 0$. Since $\text{Supp}(q_*E)$ contains all ψ -exceptional divisors and $\text{Supp}(G)$ contains all ϕ -exceptional divisors, $\text{Supp}(G + p_*E)$ contains all $(\phi \circ \psi)$ -exceptional divisors. $\square$

Solution to (8.6). Let $\phi : X \dashrightarrow Y$ be a log terminal model for (X, Δ) over Z and $\zeta : Y \rightarrow Z$ be the corresponding morphism. We must show that if $\|\Delta - \Delta_0\| \ll 1$, then ϕ is a log terminal model for (X, Δ) over U . Replacing Δ by an appropriate ($\mathbb{R}$ -linearly equivalent over U) divisor, we may assume that $\Delta = A + B$ where A is ample. By the Cone Theorem (1.7), there are finitely many extremal rays R_i over U corresponding to curves $C_i \subset Y$ such that if $K_Y + \phi_*\Delta$ is not nef over U , then $(K_Y + \phi_*\Delta) \cdot C_i < 0$. Since $(K_Y + \phi_*\Delta) \sim_{\mathbb{R}, U} \phi_*(\Delta - \Delta_0) + \zeta^*H$ and $\|\Delta - \Delta_0\| \ll 1$, we may assume that $\zeta^*H \cdot C_i = 0$ and hence that the R_i are extremal rays over Z . By assumption $K_Y + \phi_*\Delta$ is nef over Z and hence it is nef over U . $\square$

REFERENCES

- [BCHM06] C. Birkar, P. Cascini, C. D. Hacon, J. McKernan, *Existence of minimal models for varieties of log general type*. ArXiv:math.AG/0610203.
- [Eisenbud95] D. Eisenbud, *Commutative algebra. With a view toward algebraic geometry*. Graduate Texts in Mathematics, 150. Springer-Verlag, New York, 1995. xvi+785 pp.
- [Fujino07] O. Fujino, *Special termination and reduction to pl flips*, in: Flips for 3-folds and 4-folds, A. Corti editor. Oxford lecture series in mathematics and its applications, **35**.
- [FM00] O. Fujino, S. Mori, *A canonical bundle formula*. J. Differential Geom. **56** (2000), no. 1, 167–188.
- [HM07] C. D. Hacon, J. McKernan, *Extension Theorems and the existence of Flips*, in: Flips for 3-folds and 4-folds, A. Corti editor. Oxford lecture series in mathematics and its applications, **35**.
- [HM08] C. D. Hacon, J. McKernan, *Existence of minimal models for varieties of log general type II: pl-flips*. arXiv:0808.1929
- [Kollár93] J. Kollár, *Effective base point freeness*. Math. Ann. **296** (1993), no. 4, 595–605.
- [KM98] J. Kollár, S. Mori, *Birational geometry of algebraic varieties*. With the collaboration of C. H. Clemens and A. Corti. Translated from the 1998 Japanese original. Cambridge Tracts in Mathematics, 134. Cambridge University Press, Cambridge, 1998.
- [Lazarsfeld04] R. Lazarsfeld, *Positivity in algebraic geometry*. Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge. A Series of Modern Surveys in Mathematics [Results in Mathematics and Related Areas. 3rd Series. A Series of Modern Surveys in Mathematics], 48–49. Springer-Verlag, Berlin, 2004.
- [Mori88] S. Mori, *Flip theorem and the existence of minimal models for 3-folds*. J. Amer. Math. Soc. **1** (1988), no. 1, 117–253.

- [Nakayama04] N. Nakayama, *Zariski-decomposition and abundance*. MSJ Memoirs, 14. Mathematical Society of Japan, Tokyo, 2004. xiv+277 pp.
- [Shokurov03] V. V. Shokurov, *Prelimiting flips*. Tr. Mat. Inst. Steklova **240** (2003), Biratsion. Geom. Linein. Sist. Konechno Porozhdennye Algebry, 82–219; translation in Proc. Steklov Inst. Math. 2003, no. 1 (240), 75–213
- [Siu06] Y.-T. Siu, *A General Non-Vanishing Theorem and an Analytic Proof of the Finite Generation of the Canonical Ring*. math.AG/0610740
- [Szabó94] E. Szabó, *Divisorial log terminal singularities*. J. Math. Sci. Univ. Tokyo 1 (1994), no. 3, 631–639.

DEPARTMENT OF MATHEMATICS, UNIVERSITY OF UTAH, 155 SOUTH 1400 EAST, JWB 233,
SALT LAKE CITY, UT 84112, USA
E-mail address: `hacon@math.utah.edu`